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Integrated circuit, system for and method of forming an integrated circuit

Abstract

An integrated circuit structure includes a first and second power rail on a first level, a first and second set of conductive structures on a second level and a first, second and third conductive structure on a third level. The first set of conductive structures is over the first power rail. The second set of conductive structures is over the second power rail. The first conductive structure overlaps a first conductive structure of the first set of conductive structures and a first conductive structure of the second set of conductive structures. The second conductive structure overlaps a second conductive structure of the first set of conductive structures and a second conductive structure of the second set of conductive structures. The third conductive structure overlaps a third conductive structure of the first set of conductive structures and a third conductive structure of the second set of conductive structures.

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Background/Summary

PRIORITY CLAIM (1) The present application is a continuation of U.S. application Ser. No. 18/354,423, filed Jul. 18, 2023, now U.S. Pat. No. 12,073,170, issued Aug. 27, 2024, which is a continuation of U.S. application Ser. No. 17/106,876, filed Nov. 30, 2020, now U.S. Pat. No. 11,704,465, issued Jul. 18, 2023, which is a continuation of U.S. application Ser. No. 15/643,825, filed Jul. 7, 2017, now U.S. Pat. No. 10,854,499, issued Dec. 1, 2020, which claims the benefit of U.S. Provisional Application No. 62/427,749, filed Nov. 29, 2016, which are herein incorporated by reference in their entireties.

BACKGROUND

- (1) In many integrated circuits (ICs), power rails are used to distribute power to functional circuit elements formed in a substrate. Power is often delivered to power rails using metal layers between the power rails and power straps at a level above the level of the power rails.
 - (2) The resistance of an IC structure including such metal layers can affect the efficiency of power delivery, heat generation, and susceptibility to electromigration (EM). Routing of the metal layers can also impact the routing of additional electrical connections to the functional circuit elements.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1A is a top view of an IC structure, in accordance with some embodiments.
- (3) FIG. 1B is a top view of another IC structure, in accordance with some embodiments.
- (4) FIGS. 2A, 2B and 2C are diagrams of an IC structure, in accordance with some embodiments.
- (5) FIGS. 3A, 3B and 3C are diagrams of another IC structure, in accordance with some embodiments.
- (6) FIG. 4A is a diagram of a layout design, in accordance with some embodiments.
- (7) FIG. 4B is a diagram of a layout design, in accordance with some embodiments.
- (8) FIG. 5A is a diagram of a layout design, in accordance with some embodiments.
- (9) FIG. 5B is a diagram of a layout design, in accordance with some embodiments.
- (10) FIG. 6 is a diagram of a layout design, in accordance with some embodiments.
- (11) FIG. 7 is a diagram of a layout design, in accordance with some embodiments.
- (12) FIG. 8 is a diagram of a layout design, in accordance with some embodiments.
- (13) FIG. 9 is a diagram of a layout design, in accordance with some embodiments.
- (14) FIG. 10 is a diagram of a layout design, in accordance with some embodiments.
- (15) FIG. 11 is a diagram of a layout design, in accordance with some embodiments.
- (16) FIG. 12 is a diagram of a layout design, in accordance with some embodiments.
- (17) FIG. 13 is a diagram of a layout design, in accordance with some embodiments.
- (18) FIG. 14 is a diagram of a layout design, in accordance with some embodiments.
- (19) FIG. 15 is a diagram of a layout design, in accordance with some embodiments.
- (20) FIG. 16 is a diagram of a layout design, in accordance with some embodiments.
- (21) FIG. 17 is a diagram of a layout design, in accordance with some embodiments.

(22) FIG. **18** is a diagram of a layout design, in accordance with some embodiments.

(23) FIG. **19A** is a flowchart of a method of forming an IC structure, in accordance with some embodiments.

(24) FIG. **19B** is a flowchart of a method of manufacturing an IC, in accordance with some embodiments.

(25) FIG. **20** is a block diagram of a system of designing an IC layout design, in accordance with some embodiments.

DETAILED DESCRIPTION

(26) The following disclosure provides different embodiments, or examples, for implementing features of the provided subject matter. Specific examples of components, materials, values, steps, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not limiting. Other components, materials, values, steps, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(27) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(28) In accordance with some embodiments, an IC structure includes a set of rails extending in a first direction and being located at a first level, a first set of conductive structures extending in a second direction and being located at a second level different than the first level, and a first set of vias between the set of rails and the first set of conductive structures. The first set of conductive structures overlap the set of rails. The first set of vias electrically couple the set of rails to the first set of conductive structures.

(29) In some embodiments, the set of rails corresponds to a power rail configured to supply voltages VDD/VSS to other portions of the IC. In some embodiments, the first set of conductive structures corresponds to a stub power structure located on a metal 1 (M1) layer of an IC. In some embodiments, the stub power structure is electrically coupled to higher layers (e.g., metal 2 (M2) or metal 3 (M3)) within ICs. In some embodiments, M2 or M3 layers are configured as power straps. Other configurations of via layers or metal layers are within the scope of the present disclosure.

(30) In some embodiments, the stub power structure is configured to supply VDD/VSS to other portions of the IC. In some embodiments, in comparison with other approaches, the stub power structure provides more metal routing areas and via access points resulting in more electrical connections to other portions of the IC and better electromigration performance.

(31) FIG. **1A** is a top view of a portion of an IC structure **100**, in accordance with some embodiments. IC structure **100** includes a first rail **102a**, a second rail **102b**, and a third rail **102c** (hereinafter referred to as “a set of rails **120**”) each extending in a first direction X and separated from one another in a second direction Y. IC structure **100** also includes a first set of vias **125** and a first set of conductive structures **130** arranged in rows and columns. The first set of vias **125** includes one or more of vias **104a**, . . . , **104f**, **108a**, . . . , **108f**, or **112a**, . . . , **112f**. For ease of illustration, the arrow identifying the first set of vias **125** in FIGS. **1A-1B**, **2A** & **3A** points to vias

104c, 104d and 104e. However, the first set of vias **125** also refers to one or more members not identified in FIGS. **1A-1B, 2A & 3A** (e.g., one or more of vias **104a, 104b, 104f, 108a, . . . , 108f, or 112a, . . . , 112f**). The first set of conductive structures **130** includes one or more of conductive structures **106a, . . . , 106f, 110a, . . . , 110f, or 114a, . . . , 114f**. For ease of illustration, the arrow identifying the first set of conductive structures **130** in FIGS. **1A-1B, 2A & 3A** points to conductive structures **106a, 106b and 106c**. However, the first set of conductive structures **130** also refers to one or more members not identified in FIGS. **1A-1B, 2A & 3A** (e.g., one or more of conductive structures **106d, . . . , 106f, 110a, . . . , 110f, or 114a, . . . , 114f**). Three rows and six columns are used for illustration. A different number of rows or columns is within the contemplated scope of the present disclosure.

(32) Each row of vias of the first set of vias **125** is directly over and directly coupled with a rail of the set of rails **120**. Each row of conductive structures of the first set of conductive structures **130** is directly over and directly coupled with a corresponding via of the first set of vias **125**. Each rail of the set of rails **120** is electrically coupled with conductive structures of the first set of conductive structures **130** by corresponding vias of the first set of vias **125**.

(33) In an exemplary manner, first rail **102a** is electrically coupled with conductive structures **106a, 106b, . . . , 106f** by corresponding vias **104a, 104b, . . . , 104f**. Similarly, second and third rails **102b, 102c** are electrically coupled with corresponding conductive structures **110a, 110b, . . . , 110f** and **114a, 114b, . . . , 114f** by corresponding vias **108a, 108b, . . . , 108f** and **112a, 112b, . . . , 112f**. In some embodiments, integrated circuit structure **100** is configured to supply a first voltage supply **VDD**, a second supply voltage **VSS** or a current to at least a portion of an IC (not shown).

(34) Second rail **102b** is between first rail **102a** and third rail **102c**. First rail **102a** is separated from second rail **102b** in the second direction **Y** by a distance **D.sub.1a**. Second rail **102b** is separated from third rail **102c** in the second direction **Y** by a distance **D.sub.1b**. In some embodiments, distance **D.sub.1a** is the same as distance **D.sub.1b**. In some embodiments, distance **D.sub.1a** differs from distance **D.sub.1b**.

(35) First rail **102a** or third rail **102c** is configured to supply a first supply voltage **VDD** to other elements of an IC (not shown). Second rail **102b** is configured to supply a second supply voltage **VSS** to other elements of an IC (not shown). Other configurations for the first rail **102a**, second rail **102b** or third rail **102c** are within the contemplated scope of the present disclosure. In some embodiments, first rail **102a** or third rail **102c** is configured to supply the second supply voltage **VSS** to other elements of an IC (not shown), and second rail **102b** is configured to supply the first supply voltage **VDD** to other elements of an IC (not shown). In some embodiments, first rail **102a**, second rail **102b** or third rail **102c** is referred to as a power rail. Second rail **102b** alternates with first rail **102a** or third rail **102c** to supply corresponding second voltage supply **VSS** or first voltage supply **VDD**.

(36) Each rail of the set of rails **120** has a corresponding width **W.sub.1** in the second direction **Y**. In some embodiments, each width **W.sub.1** of the rails **120** is uniform. In some embodiments, at least one width **W** of a rail differs from at least one width **W.sub.1** of another rail.

(37) Each of vias **104a, 104b, . . . , 104f** is directly over and directly coupled to first rail **102a**. Similarly, each of vias **108a, 108b, . . . , 108f** and **112a, 112b, . . . , 112f** is directly over and directly coupled to corresponding second rail **102b** and third rail **102c**.

(38) A pitch **P1** between all adjacent vias of the first set of vias **125** coupled with a same rail is uniform. In some embodiments, the pitch **P1** between a pair of adjacent vias coupled with the same rail differs from the pitch **P1** between another pair of adjacent vias coupled with the same rail. For example, in these embodiments, the pitch **P1** between via **104a** and via **104b** of rail **102a** differs from the pitch **P1** between via **104b** and via **104c** of rail **102a**. In some embodiments, the pitch **P1** between a pair of adjacent vias coupled with a rail differs from the pitch **P1** between another pair of adjacent vias coupled with another rail. For example, in these embodiments, the pitch **P1** between via **104a** and via **104b** of first rail **102a** differs from the pitch **P1** between via **108b** and via **108c** of

second rail **102b**.

(39) The first set of conductive structures **130** extends in the second direction Y, and is also referred to as stubs. Each of conductive structures **106a**, **106b**, . . . , **106f** overlap the first rail **102a**.

Similarly, each of conductive structures **110a**, **110b**, . . . , **110f** and **114a**, **114b**, . . . , **114f** overlap the corresponding second and third rail **102b**, **102c**.

(40) Each of vias **104a**, **104b**, . . . , **104f** is located where corresponding conductive structures **106a**, **106b**, . . . , **106f** overlap the first rail **102a**. Similarly, each of vias **108a**, **108b**, . . . , **108f** and **112a**, **112b**, . . . , **112f** is located where corresponding conductive structures **110a**, **110b**, . . . , **110f** and **114a**, **114b**, . . . , **114f** overlap the corresponding second and third rails **102b**, **102c**.

(41) A distance D.sub.2 between all adjacent conductive structures of the first set of conductive structures **130** coupled with a same rail is uniform. In some embodiments, the distance D.sub.2 between a pair of adjacent conductive structures of the first set of conductive structures **130** coupled with the same rail differs from the distance D.sub.2 between another pair of adjacent conductive structures of the first set of conductive structures **130** coupled with the same rail. For example, in these embodiments, the distance D.sub.2 between conductive structure **106a** and conductive structure **106b** of rail **102a** differs from the distance D.sub.2 between conductive structure **106b** and conductive structure **106c** of rail **102a**.

(42) In some embodiments, the distance D.sub.2 between a pair of adjacent conductive structures of the first set of conductive structures **130** coupled with a rail differs from the distance D.sub.2 between another pair of adjacent conductive structures of the first set of conductive structures **130** coupled with another rail. For example, in these embodiments, the distance D.sub.2 between conductive structure **106a** and conductive structure **106b** of first rail **102a** differs from the distance D.sub.2 between conductive structure **108b** and conductive structure **108c** of second rail **102b**.

(43) Each conductive structure of the first set of conductive structures **130** has a corresponding width W.sub.2 in the second direction Y. In some embodiments, each width W.sub.2 of the first set of conductive structures **130** is uniform. In some embodiments, at least one width W.sub.2 of a conductive structure of the first set of conductive structures **130** differs from at least one width W.sub.2 of another conductive structure of the first set of conductive structures **130**.

(44) In some embodiments, at least one width W.sub.2 of a conductive structure of the first set of conductive structures **130** is the same as at least one width W.sub.1 of a rail of the set of rails **120**. In some embodiments, at least one width W.sub.2 of a conductive structure of the first set of conductive structures **130** differs from at least one width W.sub.1 of a rail of the set of rails **120**.

(45) A conductive structure of conductive structures **106a**, **106b**, . . . , **106f** is separated from a corresponding conductive structure of conductive structures **110a**, **110b**, . . . , **110f** in the second direction Y by a distance D.sub.3a. A conductive structure of conductive structures **110a**, **110b**, . . . , **110f** is separated from a corresponding conductive structure of conductive structures **114a**, **114b**, . . . , **114f** in the second direction Y by a distance D.sub.3b. In some embodiments, distance D.sub.3a is the same as distance D.sub.3b. In some embodiments, distance D.sub.3a differs from distance D.sub.3b. In some embodiments, at least one distance of distances D.sub.1a or D.sub.1b is the same as at least one distance of distances D.sub.3a or D.sub.3b. In some embodiments, at least one distance of distances D.sub.1a or D.sub.1b differs from at least one distance of distances D.sub.3a or D.sub.3b.

(46) In some embodiments, by not coupling one or more of conductive structures **106a**, . . . , **106f**, **110a**, . . . , **110f**, **114a**, . . . , **114f** to other conductive structures of conductive structures **106a**, . . . , **106f**, **110a**, . . . , **110f**, **114a**, . . . , **114f** on the same metal layer, the space between two or more of conductive structures **106a**, . . . , **106f**, **110a**, . . . , **110f**, **114a**, . . . , **114f** can be utilized for additional routing resources on the same metal layer, and the additional routing resources can provide additional via access points to other metal layers resulting in IC structure **100** having a reduced size compared with other approaches.

(47) FIG. 1B is a top view of a portion of another IC structure **100'**, in accordance with some

embodiments. Components that are the same or similar to those in FIG. 1A are given the same reference numbers, and detailed description thereof is thus omitted.

(48) In comparison with IC structure **100** of FIG. 1A, IC structure **100'** includes fewer vias of the first set of vias **125** and fewer conductive structures of the first set of conductive structures **130**. For example, IC structure **100'** does not include vias **104b**, **104d** and **104f**, vias **108a**, **108c** and **108e**, vias **112b**, **112d** and **112f**, conductive structures **106b**, **106d** and **106f**, conductive structures **110a**, **110c** and **110e** and conductive structures **114b**, **114d** and **114f**.

(49) Each row of vias of the first set of vias **125** is offset from an adjacent row of vias by an offset distance D_{off}. For example, vias **104a**, **104c** and **104e** alternate with vias **108b**, **108d** and **108f** in the first direction X. Similarly, vias **112a**, **112c** and **112e** alternate with vias **108b**, **108d** and **108f** in the first direction X.

(50) Each row of conductive structures of the first set of conductive structures **130** is offset from an adjacent row of conductive structures by offset distance D_{sub.off}. For example, conductive structures **106a**, **106c** and **106e** alternate with conductive structures **110b**, **110d** and **110f** in the first direction X. Similarly, conductive structures **114a**, **114c** and **114e** alternate with conductive structures **110b**, **110d** and **110f** in the first direction X.

(51) By including fewer vias of the first set of vias **125** in IC structure **100'**, a pitch P₂ between adjacent vias of a same rail is increased when compared with IC structure **100**. A pitch P₂ between all adjacent vias of the first set of vias **125** coupled with a same rail is uniform. In some embodiments, the pitch P₂ between a pair of adjacent vias coupled with the same rail differs from the pitch P₂ between another pair of adjacent vias coupled with the same rail. For example, in these embodiments, the pitch P₂ between via **104a** and via **104c** of rail **102a** differs from the pitch P₂ between via **104c** and via **104e** of rail **102a**. In some embodiments, the pitch P₂ between a pair of adjacent vias coupled with a rail differs from the pitch P₂ between another pair of adjacent vias coupled with another rail. For example, in these embodiments, the pitch P₂ between via **104a** and via **104c** of first rail **102a** differs from the pitch P₂ between via **108b** and via **108d** of second rail **102b**.

(52) By including fewer conductive structures of the first set of conductive structures **130** in IC structure **100'**, a distance D_{sub.2'} between adjacent conductive structures of a same rail is increased when compared with IC structure **100**. A distance D_{sub.2'} between all adjacent conductive structures of the first set of conductive structures **130** coupled with a same rail is uniform. In some embodiments, the distance D_{sub.2'} between a pair of adjacent conductive structures of the first set of conductive structures **130** coupled with the same rail differs from the distance D_{sub.2'} between another pair of adjacent conductive structures of the first set of conductive structures **130** coupled with the same rail. For example, in these embodiments, the distance D_{sub.2'} between conductive structure **106a** and conductive structure **106c** of rail **102a** differs from the distance D_{sub.2'} between conductive structure **106c** and conductive structure **106e** of rail **102a**.

(53) In some embodiments, the distance D_{sub.2'} between a pair of adjacent conductive structures of the first set of conductive structures **130** coupled with a rail differs from the distance D_{sub.2'} between another pair of adjacent conductive structures of the first set of conductive structures **130** coupled with another rail. For example, in these embodiments, the distance D_{sub.2'} between conductive structure **106a** and conductive structure **106c** of first rail **102a** differs from the distance D_{sub.2'} between conductive structure **108b** and conductive structure **108d** of second rail **102b**.

(54) In some embodiments, by not coupling one or more of conductive structures **106a**, **106c**, **106e**, **110b**, **110d**, **110f**, **114a**, **114c** or **114e** to other conductive structures of conductive structures **106a**, **106c**, **106e**, **110b**, **110d**, **110f**, **114a**, **114c** or **114e** on the same metal layer, the space between two or more of conductive structures **106a**, **106c**, **106e**, **110b**, **110d**, **110f**, **114a**, **114c** or **114e** can be utilized for additional routing resources on the same metal layer, and the additional routing resources can provide additional via access points to other metal layers resulting in IC structure

100' having a reduced size compared with other approaches.

(55) FIGS. 2A, 2B and 2C are diagrams of an IC structure **200**, in accordance with some embodiments. FIG. 2A is a top view of IC structure **200**, FIG. 2B is a cross-sectional view of IC structure **200** as intersected by plane A-A', and FIG. 2C is a cross-sectional view of IC structure **200** as intersected by plane B-B'.

(56) Components that are the same or similar to those in FIGS. 1A-1B, 3A-3C (shown below), 4A-4B (shown below), 5A-5 (shown below) are given the same reference numbers, and detailed description thereof is thus omitted.

(57) IC structure **200** includes IC structure **100** and a second set of vias **225** arranged in the same rows and columns of FIG. 1A. The second set of vias **225** includes one or more of vias **202a**, **202b**, . . . , **202f**, vias **204a**, **204b**, . . . , **204f** or vias **206a**, **206b**, . . . , **206f**. For ease of illustration, the arrow identifying the second set of vias **225** in FIGS. 2A & 3A points to vias **202a**, **202b** and **202c**. However, the second set of vias **225** also refers to one or more members not identified in FIGS. 2A & 3A (e.g., one or more of vias **202d**, . . . , **202f**, vias **204a**, **204b**, . . . , **204f** or vias **206a**, **206b**, . . . , **206f**).

(58) IC structure **200** further includes a second set of conductive structures **230** arranged in rows, a third set of conductive structures **240** arranged in columns and a third set of vias **235**.

(59) The second set of conductive structures **230** includes one or more of a first conductive structure **208a**, a second conductive structure **208b** or a third conductive structure **208c**.

(60) The third set of conductive structures **240** includes one or more of conductive structures **220a**, **220b**, . . . , **220f**. For ease of illustration, the arrow identifying the third set of conductive structures **240** in FIGS. 2A & 3A points to conductive structures **220b**, **220c** and **220d**. However, the third set of conductive structures **240** also refers to one or more members not identified in FIGS. 2A & 3A (e.g., one or more of conductive structures **220a**, **220e**, **220f**).

(61) The third set of vias **235** includes one or more of vias **210a**, **210b**, **210c**, vias **212a**, **212b**, **212c** or vias **214a**, **214b**, **214c**. For ease of illustration, the arrow identifying the third set of vias **235** in FIGS. 2A & 3A points to vias **210b** and **210c**. However, the third set of vias **235** also refers to one or more members not identified in FIGS. 2A & 3A (e.g., one or more of vias **210a**, vias **212a**, **212b**, **212c** or vias **214a**, **214b**, **214c**).

(62) Each row of vias of the second set of vias **225** is directly over and directly coupled with a corresponding conductive structure of the first set of conductive structures **130**. In an exemplary manner, each of vias **202a**, **202b**, . . . , **202f** is directly over and directly coupled to corresponding conductive structures **106a**, **106b**, . . . , **106f**. Similarly, each of vias **204a**, **204b**, . . . , **204f**, **206a**, **206b**, . . . , **206f**, is directly over and directly coupled to corresponding conductive structures **110a**, **110b**, . . . , **110f**, **114a**, **114b**, . . . , **114f**.

(63) Each row of conductive structures of the second set of conductive structures **230** is directly over and directly coupled with a corresponding row of vias of the second set of vias **225**. In an exemplary manner, each of vias **202a**, **202b**, . . . , **202f** is directly below and directly coupled to first conductive structure **208a**. Similarly, each of vias **204a**, **204b**, . . . , **204f**, and vias **206a**, **206b**, . . . , **206f** is directly below and directly coupled to corresponding second and third conductive structure **208b**, **208c**.

(64) Each conductive structure of the second set of conductive structures **230** is electrically coupled with conductive structures of the first set of conductive structures **130** by corresponding vias of the second set of vias **225**. In an exemplary manner, first conductive structure **208a** is electrically coupled to conductive structures **106a**, **106b**, . . . , **106f** by corresponding vias **202a**, **202b**, . . . , **202f**. Similarly, second and third conductive structures **208b**, **208c** are electrically coupled to corresponding conductive structures **110a**, **110b**, . . . , **110f**, and **114a**, **114b**, . . . , **114f** by corresponding vias **204a**, **204b**, . . . , **204f** and vias **206a**, **206b**, . . . , **206f**.

(65) The second set of vias **225** has the same pitch **P1** as the pitch **P1** for the first set of vias **125**. In some embodiments, at least one pitch **P1** between a pair of adjacent vias of the second set of vias

225 differs from at least one pitch **P1** between a pair of adjacent vias of the first set of vias **125**. For example, in these embodiments, the pitch **P1** between via **104a** and via **104b** of rail **102a** differs from the pitch **P1** between via **202a** and via **202b** of first conductive structure **208a**.

(66) Each of first conductive structure **208a**, second conductive structure **208b** and third conductive structure **208c** extend in the first direction **X** and are separated from one another in the second direction **Y**. Second conductive structure **208b** is between first conductive structure **208a** and third conductive structure **208c**. First conductive structure **208a** is separated from second conductive structure **208b** in the second direction **Y** by a distance $D_{sub.1a'}$. Second conductive structure **208b** is separated from third conductive structure **208c** in the second direction **Y** by a distance $D_{sub.1b'}$. In some embodiments, distance $D_{sub.1a'}$ is the same as distance $D_{sub.1b'}$. In some embodiments, distance $D_{sub.1a'}$ differs from distance $D_{sub.1b'}$.

(67) Each conductive structure of the second set of conductive structures **230** has a corresponding width $W_{sub.1'}$ in the second direction **Y**. In some embodiments, each width $W_{sub.1'}$ of the second set of conductive structures **230** is uniform. In some embodiments, at least one width $W_{sub.1'}$ of the second set of conductive structures **230** differs from at least one width $W_{sub.1'}$ of another conductive structure of the second set of conductive structures **230**.

(68) Each of vias **210a**, **210b**, **210c** is directly over and directly coupled to first conductive structure **208a**. Similarly, vias **212a**, **212b**, **212c** and vias **214a**, **214b**, **214c** are directly over and directly coupled to corresponding second and third conductive structures **208b**, **208c**.

(69) Each of vias **210a**, **210b**, **210c** is directly below and directly coupled to a corresponding conductive structure of conductive structures **220a**, **220c**, **220e**. Similarly, each of vias **212a**, **212b**, **212c**, **214a**, **214b**, **214c** is directly below and directly coupled to a corresponding conductive structure of conductive structures **220b**, **220d**, **220f**, **220a**, **220c**, **220e**.

(70) An arrangement of the third set of vias **235** is similar to the arrangement of the first set of vias **125** shown in FIG. 1B. For example, each of vias **210a**, **210b**, **210c**, **212a**, **212b**, **212c**, **214a**, **214b**, **214c** is located at a same corresponding location in the x-y plane as corresponding vias **104a**, **104c**, **104e**, **108b**, **108d**, **108f**, **112a**, **112c**, **112e** shown in FIG. 1B. In some embodiments, a location of a via of the third set of vias **235** differs from a location of a corresponding via of the first set of vias **125**. Furthermore, a pitch **P1** between adjacent vias of the third set of vias **235** is the same as pitches **P1** between adjacent vias of the first set of vias **125**. In some embodiments, a pitch **P1** between adjacent vias of the third set of vias **235** differs from a pitch **P1** between adjacent vias of the first set of vias **125**.

(71) Each of conductive structures **220a**, **220b**, . . . , **220f** extend in the second direction **Y**. The fourth set of conductive structures **240** is also referred to as straps. In some embodiments, a strap is a conductive structure that overlaps and is electrically coupled to two or more underlying, conductive structures. Each of conductive structures **220a**, **220b**, . . . , **220f** overlap the first conductive structure **208a**, second conductive structure **208b**, and third conductive structure **208c**.

(72) A distance $D_{sub.3'}$ between all adjacent conductive structures of the third set of conductive structures **240** is the same as the distance $D_{sub.2}$ (shown in FIG. 1A) between all adjacent conductive structures of the first set of conductive structures **130**. In some embodiments, the distance $D_{sub.3'}$ between a pair of adjacent conductive structures of the third set of conductive structures **240** differs from the distance $D_{sub.3'}$ between another pair of adjacent conductive structures of the third set of conductive structures **240**. In some embodiments, the distance $D_{sub.3'}$ between a pair of adjacent conductive structures of the third set of conductive structures **240** differs from the distance $D_{sub.2}$ (shown in FIG. 1A) between a pair of adjacent conductive structures of the first set of conductive structures **130** coupled to rail **102a**.

(73) Conductive structure **220a** is electrically coupled to the first conductive structure **208a** by via **210a**, and electrically coupled to the third conductive structure **208c** by via **214a**. Via **210a** is located where conductive structure **220a** overlaps first conductive structure **208a**. Via **214a** is located where conductive structure **220a** overlaps third conductive structure **208c**.

(74) Conductive structure **220a** is configured to supply first voltage supply VDD. In some embodiments, conductive structure **220a** is configured to supply second voltage supply VSS.

(75) Conductive structure **220b** is electrically coupled to the second conductive structure **208b** by via **212a**. Via **212a** is located where conductive structure **220b** overlaps second conductive structure **208b**.

(76) Conductive structure **220c** is electrically coupled to the first conductive structure **208a** by via **210b**, and electrically coupled to the third conductive structure **208c** by via **214b**. Via **210b** is located where conductive structure **220c** overlaps first conductive structure **208a**. Via **214b** is located where conductive structure **220c** overlaps third conductive structure **208c**.

(77) Conductive structure **220d** is electrically coupled to the second conductive structure **208b** by via **212b**. Via **212b** is located where conductive structure **220d** overlaps second conductive structure **208b**.

(78) Conductive structure **220e** is electrically coupled to the first conductive structure **208a** by via **210c**, and electrically coupled to the third conductive structure **208c** by via **214c**. Via **210c** is located where conductive structure **220e** overlaps first conductive structure **208a**. Via **214c** is located where conductive structure **220e** overlaps third conductive structure **208c**.

(79) Conductive structure **220f** is electrically coupled to the second conductive structure **208b** by via **212c**. Via **212c** is located where conductive structure **220f** overlaps second conductive structure **208b**.

(80) Conductive structure **220a**, **220c** or **220e** is configured to supply first voltage supply VDD. In some embodiments, conductive structure **220a**, **220c** or **220e** is configured to supply second voltage supply VSS. Conductive structure **220b**, **220d** or **220f** is configured to supply second voltage supply VSS. In some embodiments, conductive structure **220b**, **220d** or **220f** is configured to supply first voltage supply VDD. In some embodiments, conductive structures **220a**, **220b**, . . . , **220f** are referred to as a power strap.

(81) As shown in FIGS. 2B-2C (in part), first rail **102a**, second rail **102b** and third rail **102c** are positioned at a metal zero (M0) layer of IC structure **200**. Vias **104a**, **104b**, . . . , **104f**, vias **108a**, **108b**, . . . , **108f** and vias **112a**, **112b**, . . . , **112f** are positioned at a via zero (V0) layer of IC structure **200**. Conductive structures **106a**, **106b**, . . . , **106f**, conductive structures **110a**, **110b**, . . . , **110f**, and conductive structures **114a**, **114b**, . . . , **114f** are positioned at a metal one (M1) layer of IC structure **200**. Each of the conductive structures that are positioned at the M1 layer extend in the same direction. Vias **202a**, **202b**, . . . , **202f**, vias **204a**, **204b**, . . . , **204f** and vias **206a**, **206b**, . . . , **206f** are positioned at a via one (V1) layer of IC structure **200**. First conductive structure **208a**, second conductive structure **208b** and third conductive structure **208c** are positioned at a metal two (M2) layer of IC structure **200**. Vias **210a**, **210b**, **210c**, vias **212a**, **210b**, **210c** and vias **214a**, **214b**, **214c** are positioned at a via two (V2) layer of IC structure **200**. Conductive structures **220a**, **220b**, **220f** are positioned at a metal three (M3) layer of IC structure **200**. Other configurations of via layers or metal layers are within the scope of the present disclosure.

(82) M1 layer is separated from M0 layer by V0 layer. In some embodiments, one or more metal layers (not shown) separate M0 layer and M1 layer. M2 layer is separated from M1 layer by V1 layer. In some embodiments, one or more metal layers (not shown) separate M1 layer and M2 layer. M3 layer is separated from M2 layer by V2 layer. In some embodiments, one or more metal layers (not shown) separate M2 layer and M3 layer. In some embodiments, each of the conductive structures of IC structure **100**, **200** located in the M1 layer extend in the same direction. Other configurations of via layers or metal layers are within the scope of the present disclosure.

(83) A conductive structure of the first set of conductive structures **130** or the third set of conductive structures **240** is parallel to another conductive structure of the first set of conductive structures **130** or the third set of conductive structures **240**. In some embodiments, a conductive structure of the first set of conductive structures **130** or the third set of conductive structures **240** is not parallel to another conductive structure of the first set of conductive structures **130** or the third

set of conductive structures **240**.

(84) A member of the set of rails **120** or the second set of conductive structures **230** is parallel to another member of the set of rails **120** or the second set of conductive structures **230**. In some embodiments, a member of the set of rails **120** or the second set of conductive structures **230** is not parallel to another member of the set of rails **120** or the second set of conductive structures **230**.

(85) In some embodiments, at least one member of the set of rails **120**, the first set of conductive structures **130**, the second set of conductive structures **230** or the third set of conductive structures **240** is a conductive material including copper, aluminum, alloys thereof or other suitable conductive materials, that is formed in one or more metallization layers by one or more of a physical vapor deposition process, a chemical vapor deposition process, a plating process, or other suitable processes.

(86) In some embodiments, at least one via of the first set of vias **125**, the second set of vias **225** or the third set of vias **235** is a metal line, a via, a through silicon via (TSV), an inter-level via (ILV), a slot via, an array of vias, or another suitable conductive line. In some embodiments, at least one via of the first set of vias **125**, the second set of vias **225** or the third set of vias **235** includes copper, aluminum, nickel, titanium, tungsten, cobalt, carbon, alloys thereof or another suitable conductive material, that is formed in one or more metallization layers by one or more of a physical vapor deposition process, a chemical vapor deposition process, a plating process, or other suitable processes. In some embodiments, at least one via of the first set of vias **125**, the second set of vias **225** or the third set of vias **235** includes one or more conductive line portions.

(87) In some embodiments, by not coupling one or more of conductive structures **106a**, . . . , **106f**, **110a**, . . . , **110f**, **114a**, . . . , **114f** to other conductive structures of conductive structures **106a**, . . . **106f**, **110a**, . . . , **110f**, **114a**, . . . , **114f** on the same metal layer (e.g., M1), the space between two or more of conductive structures **106a**, . . . , **106f**, **110a**, . . . , **110f**, **114a**, . . . , **114f** can be utilized for additional M1 routing resources, and the additional routing resources can provide additional via access points to other metal layers resulting in IC structure **200** having a reduced size compared with other approaches. Other configurations of via layers or metal layers are within the scope of the present disclosure.

(88) FIGS. 3A, 3B and 3C are diagrams of another IC structure **300**, in accordance with some embodiments. FIG. 3A is a top view of IC structure **300**, FIG. 3B is a cross-sectional view of IC structure **300** as intersected by plane C-C', and FIG. 3C is a cross-sectional view of IC structure **300** as intersected by plane D-D'.

(89) IC structure **300** is a variation of IC structure **100'** (FIG. 1B) or IC structure **200** (FIG. 2A). Components that are the same or similar to those in FIGS. 1A-1B or 2A-2C are given the same reference numbers, and detailed description thereof is thus omitted.

(90) In comparison with IC structure **200** of FIG. 2A, IC structure **300** includes fewer vias of the first set of vias **125**, fewer conductive structures of the first set of conductive structures **130** and fewer vias of the second set of vias **225**. For example, IC structure **300** does not include vias **104b**, **104d**, **104f**, vias **108a**, **108c**, **108e**, vias **112b**, **112d**, **112f**, conductive structures **106b**, **106d**, **106f**, conductive structures **110a**, **110c**, **110e**, conductive structures **114b**, **114d**, **114f**, vias **202b**, **202d**, **202f**, vias **204a**, **204c**, **204e**, and vias **206b**, **206d**, **206f**.

(91) Each row of vias of the second set of vias **225** is offset from an adjacent row of vias by the offset distance D_{off}. Vias **202a**, **202c** and **202e** alternate with vias **204b**, **204d** and **204f** in the first direction X. Similarly, vias **204b**, **204d** and **204f** alternate with vias **206a**, **206c** and **206e** in the first direction X.

(92) Vias **210a**, **210c** and **210e** alternate with vias **212b**, **212d** and **212f** in the first direction X. Similarly, vias **212b**, **212d** and **212f** alternate with vias **214a**, **214c** and **214e** in the first direction X.

(93) By including fewer vias of the second set of vias **225** in IC structure **300**, a pitch P2 between adjacent vias of the second set of vias **225** corresponds to a pitch P2 between adjacent vias of the third set of vias **235**. For example, in these embodiments, the pitch P2 between via **202a** and via

202c of rail 102a is the same as the pitch P2 between via 210a and via 210b.

(94) In some embodiments, by not coupling one or more of conductive structures 106a, 106c, 106e, 110b, 110d, 110f, 114a, 114c or 114e to other conductive structures of conductive structures 106a, 106c, 106e, 110b, 110d, 110f, 114a, 114c or 114e on the same metal layer (e.g., M1), the space between two or more of conductive structures 106a, 106c, 106e, 110b, 110d, 110f, 114a, 114c or 114e can be utilized for additional M1 routing resources, and the additional M1 routing resources can provide additional via access points to other metal layers resulting in IC structure 200' having reduced size compared with other approaches. Other configurations of via layers or metal layers are within the scope of the present disclosure.

(95) FIG. 4A is a diagram of a layout design 400, in accordance with some embodiments. Layout design 400 is usable to manufacture a portion of IC structure 200 (FIGS. 2A-2C). Components that are the same or similar to those in FIGS. 1A-1B or 2A-2C are given the same reference numbers, and detailed description thereof is thus omitted. In some embodiments, one or more of the layout patterns described herein (e.g., layout design 400, 400', 500, 500' or 600-1800) is usable to prepare a set of masks, that are in turn usable for manufacturing an IC structure (e.g., IC structure 100, 100', 200 or 300). The layout design 400 is a basis to be modified to form other layout designs, such as those described herein, e.g., FIGS. 4B, 5A-5B & 6-18. Layout design 400 depicts overlapping layout patterns from various layout layers. Some layout patterns and some layout layers of layout design 400 are simplified or omitted.

(96) Layout design 400 includes a first conductive layout pattern 401a, a second conductive layout pattern 401b, a third conductive layout pattern 401c, power layout patterns 402, 404, 406, 408, 410 and 412, a first region 420, a second region 422, a third region 424 and a fourth region 426.

(97) First, second and third conductive layout patterns 401a, 401b, 401c are usable by system 2000 (FIG. 20) to form corresponding first, second and third rails 102a, 102b, 102c (FIGS. 1A-1B, 2A-2C, 3A-3C). The arrangement of the first, second and third conductive layout patterns 401a, 401b, 401c is similar to the arrangement of the first, second and third rails 102a, 102b, 102c (FIGS. 1A-1B, 2A-2C, 3A-3C).

(98) Power layout patterns 402, 404, 406, 408, 410, 412 are usable by system 2000 (FIG. 20) to form corresponding conductive structures 114a, 114b, 110a, 110b, 106a, 106b (FIGS. 1A-1B, 2A-2C, 3A-3C).

(99) A center of power layout pattern 402 or 404 is symmetric or aligned with a center of third conductive layout pattern 401c. The center of third conductive layout pattern 401c is aligned with first line 440. Power layout patterns 402 and 404 are aligned with each other in the first direction X. In some embodiments, power layout patterns 402 and 404 are not aligned with each other in the first direction X. Power layout pattern 402 is offset from power layout pattern 404 in the first direction X by a pitch P4. Power layout patterns 402, 404 are over third conductive layout pattern 401c.

(100) Power layout pattern 402 comprises a conductive feature layout pattern 402a, a cut feature layout pattern 402b, a cut feature layout pattern 402c and a via layout pattern 402d. Conductive feature layout pattern 402a is usable to manufacture conductive structure 114a, and via layout pattern 402d is usable to manufacture via 112a (FIGS. 1A-1B, 2A-2C). Via layout pattern 402d is symmetric with respect to first line 440 of third conductive layout pattern 401c. In some embodiments, the cut feature layout pattern 402b or 402c identifies a location of the conductive structure 114a (FIG. 1A-1B) that is removed during fabrication of IC structure 100, 100'. In some embodiments, the removed portion of the conductive structure 114a corresponds to a cut region (e.g., cut feature layout pattern 402b, 402c).

(101) Power layout pattern 404 is similar to power layout pattern 402 with the reference numerals increased by 2. Conductive feature layout pattern 404a is usable to manufacture conductive structure 114b, and via layout pattern 404d is usable to manufacture via 112b (FIGS. 1A-1B, 2A-2C). Via layout pattern 404d is symmetric with respect to first line 440 of third conductive layout

pattern **401c**. In some embodiments, the cut feature layout pattern **404b** or **404c** identifies a location of the conductive structure **114b** that is removed during fabrication of IC structure **100**, **100'**. In some embodiments, the removed portion of the conductive structure **114b** corresponds to a cut region (e.g., cut feature layout pattern **404b**, **404c**).

(102) Power layout patterns **406** and **408** are similar to corresponding power layout patterns **402** and **404** with the reference numerals increased by 4. A center of power layout pattern **406** or **408** is symmetric or aligned with a center of second conductive layout pattern **401b**. Power layout patterns **406** and **408** are aligned with each other in the first direction X. In some embodiments, power layout patterns **406** and **408** are not aligned with each other in the first direction X. Power layout pattern **406** is offset from power layout pattern **408** in the first direction X by pitch P4. A second line **442** is aligned with a center of second conductive layout pattern **401b**. Power layout patterns **406**, **408** are over second conductive layout pattern **401b**.

(103) Conductive feature layout patterns **406a**, **408a** are usable to manufacture corresponding conductive structures **110a**, **110b** and via layout patterns **406d**, **408d** are usable to manufacture corresponding vias **108a**, **108b** (FIGS. 1A-1B, 2A-2C). Via layout patterns **406d**, **408d** are symmetric with respect to second line **442** of second conductive layout pattern **401b**. In some embodiments, the cut feature layout pattern **406b** or **406c** identifies a location of the conductive structure **110a** that is removed during fabrication of IC structure **100**, **100'**. In some embodiments, the removed portion of the conductive structure **110a** corresponds to a cut region (e.g., cut feature layout pattern **406b**, **406c**). In some embodiments, the cut feature layout pattern **408b** or **408c** identifies a location of the conductive structure **110b** that is removed during fabrication of IC structure **100**, **100'**. In some embodiments, the removed portion of the conductive structure **110b** corresponds to a cut region (e.g., cut feature layout pattern **408b**, **408c**).

(104) Power layout patterns **410** and **412** are similar to as corresponding power layout patterns **402** and **404** with the reference numerals increased by 8. A center of power layout pattern **410** or **412** is symmetric or aligned with a center of first conductive layout pattern **401a**. Power layout patterns **410** and **412** are aligned with each other in the first direction X. In some embodiments, power layout patterns **410** and **412** are not aligned with each other in the first direction X. Power layout pattern **410** is offset from power layout pattern **412** in the first direction X by pitch P4. A third line **444** is aligned with a center of first conductive layout pattern **401a**. Power layout patterns **410**, **412** are over first conductive layout pattern **401a**.

(105) Conductive feature layout patterns **410a**, **412a** are usable to manufacture corresponding conductive structures **106a**, **106b** and via layout patterns **410d**, **412d** are usable to manufacture corresponding vias **104a**, **104b** (FIGS. 1A-1B, 2A-2C). Via layout patterns **410d**, **412d** are symmetric with respect to third line **444** of first conductive layout pattern **401a**. In some embodiments, the cut feature layout pattern **410b** or **410c** identifies a location of the conductive structure **106a** that is removed during fabrication of IC structure **100**, **100'**. In some embodiments, the removed portion of the conductive structure **106a** corresponds to a cut region (e.g., cut feature layout pattern **410b**, **410c**). In some embodiments, the cut feature layout pattern **412b** or **412c** identifies a location of the conductive structure **106b** that is removed during fabrication of IC structure **100**, **100'**. In some embodiments, the removed portion of the conductive structure **106b** corresponds to a cut region (e.g., cut feature layout pattern **412b**, **412c**).

(106) Power layout pattern **402** or **404** is separated from corresponding power layout pattern **406** or **408** in the second direction Y by distance D.sub.4b. Power layout pattern **406** or **408** is separated from corresponding power layout pattern **410** or **412** in the second direction by distance D.sub.4a.

(107) First region **420** comprises a conductive feature layout pattern **420a** and a conductive feature layout pattern **420b**.

(108) Conductive feature layout pattern **420a** or conductive feature layout pattern **420b** is coupled to other elements (not shown) of IC structure **100**, **200**. Conductive feature layout pattern **420a** and conductive feature layout pattern **420b** are between second conductive layout pattern **401b** and

third conductive layout pattern **401c**. Conductive feature layout pattern **420b** is between power layout pattern **402** and power layout pattern **406**. In some embodiments, conductive feature layout pattern **420a** is between power layout pattern **402** and power layout pattern **406**. In some embodiments, conductive feature layout pattern **420b** is aligned with conductive feature layout pattern **402a** in the second direction Y. Conductive feature layout pattern **420b** has a length $L_{sub.1}$ in the second direction Y. Conductive feature layout pattern **420b** does not overlap power layout pattern **402** or power layout pattern **406**. Conductive feature layout pattern **420a** has a first color (e.g., color A) and conductive feature layout pattern **420b** has a second color (e.g., color B) different than the first color. In some embodiments, conductive feature layout pattern **420a** and conductive feature layout pattern **420b** have a same color (e.g., color A or B) as each other. The color (e.g., color A, color B) indicates that features with the same color are to be formed on a same mask of a multiple mask set, and features with a different color are to be formed on a different mask of the multiple mask set.

(109) In some embodiments, by not coupling power layout pattern **402** to power layout pattern **406** on the same metal layer (e.g., M1), the space between power layout pattern **402** and power layout pattern **406** can be utilized to form additional routing resources, and the additional routing resources can provide additional via access points to other metal layers. For example, by positioning conductive feature layout patterns **420a** or **420b** between power layout pattern **402** and power layout pattern **406**, conductive feature layout pattern **420a** or **420b** can be utilized to form corresponding conductive structures which provide additional M1 routing resources and via access points to other metal layers, resulting in a layout design **400** that occupies less area than other designs.

(110) Second region **422** comprises a conductive feature layout pattern **422a** and a conductive feature layout pattern **422b**.

(111) Conductive feature layout pattern **422a** or conductive feature layout pattern **422b** is coupled to other elements (not shown) of IC structure **100**, **200**. Conductive feature layout pattern **422a** and conductive feature layout pattern **422b** are between second conductive layout pattern **401b** and third conductive layout pattern **401c**. Conductive feature layout pattern **422b** is between power layout pattern **404** and power layout pattern **408**. In some embodiments, conductive feature layout pattern **422a** is between power layout pattern **404** and power layout pattern **408**. In some embodiments, conductive feature layout pattern **422b** is aligned with conductive feature layout pattern **404a** in the second direction Y. Conductive feature layout pattern **422b** has a length $L_{sub.2}$ in the second direction Y. Conductive feature layout pattern **422b** does not overlap power layout pattern **404** or power layout pattern **408**. Conductive feature layout pattern **422a** has a first color (e.g., color A) and conductive feature layout pattern **422b** has a second color (e.g., color B) different than the first color. In some embodiments, conductive feature layout pattern **422a** and conductive feature layout pattern **422b** have a same color (e.g., color A or B) as each other.

(112) In some embodiments, by not coupling power layout pattern **404** to power layout pattern **408** on the same metal layer (e.g., M1), the space between power layout pattern **404** and power layout pattern **408** can be utilized to form additional routing resources, and the additional routing resources can provide additional via access points to other metal layers. For example, by positioning conductive feature layout patterns **422a** or **422b** between power layout pattern **404** and power layout pattern **408**, conductive feature layout patterns **422a** or **422b** can be utilized to form corresponding conductive structures which provide additional M1 routing resources and via access points to other metal layers, resulting in a layout design **400** that occupies less area than other designs.

(113) Third region **424** comprises a conductive feature layout pattern **424a** and a conductive feature layout pattern **424b**.

(114) Conductive feature layout pattern **424a** or conductive feature layout pattern **424b** is coupled to other elements (not shown) of IC structure **100**, **200**. Conductive feature layout pattern **424a** and

conductive feature layout pattern **424b** are between first conductive layout pattern **401a** and second conductive layout pattern **401b**. Conductive feature layout pattern **424b** is between power layout pattern **406** and power layout pattern **410**. In some embodiments, conductive feature layout pattern **424a** is between power layout pattern **406** and power layout pattern **410**. In some embodiments, conductive feature layout pattern **424b** is aligned with conductive feature layout pattern **406a** in the second direction Y. Conductive feature layout pattern **424b** has a length $L_{sub.3}$ in the second direction Y. In some embodiments, length $L_{sub.1}$ is the same as length $L_{sub.3}$. In some embodiments, length $L_{sub.1}$ differs from length $L_{sub.3}$. Conductive feature layout pattern **424b** does not overlap power layout pattern **406** or power layout pattern **410**. Conductive feature layout pattern **424a** has a first color (e.g., color A) and conductive feature layout pattern **424b** has a second color (e.g., color B) different than the first color. In some embodiments, conductive feature layout pattern **424a** and conductive feature layout pattern **424b** have a same color (e.g., color A or B) as each other.

(115) In some embodiments, by not coupling power layout pattern **406** to power layout pattern **410** on the same metal layer (e.g., M1), the space between power layout pattern **406** and power layout pattern **410** can be utilized to form additional routing resources, and the additional routing resources can provide additional via access points to other metal layers. For example, by positioning conductive feature layout patterns **424a** or **424b** between power layout pattern **406** and power layout pattern **410**, conductive feature layout patterns **424a**, **424b** can be utilized to form corresponding conductive structures which provide additional M1 routing resources and via access points to other metal layers, resulting in a layout design **400** that occupies less area than other designs.

(116) By power layout pattern **406** being symmetric with respect to second conductive layout pattern **401b**, power layout pattern **402** being symmetric with respect to third conductive layout pattern **401c**, and power layout pattern **410** being symmetric with respect to first conductive layout pattern **401a**, the length $L_{sub.1}$ of conductive feature layout pattern **420b** is less than or equal to distance $D_{sub.4b}$, and the length $L_{sub.3}$ of conductive feature layout pattern **424b** is less than or equal to distance $D_{sub.4a}$. In some embodiments, distance $D_{sub.4a}$ or $D_{sub.4b}$ is based on a height of the first region **420**, second region **422**, third region **424** or fourth region **426**. In some embodiments, a minimum value of distance $D_{sub.4a}$ or $D_{sub.4b}$ is sufficient to provide conductive feature layout pattern **420b**, **422b**, **424b** or **426b** with at least two via access points to provide a connection to upper layers (not shown).

(117) Fourth region **426** comprises a conductive feature layout pattern **426a** and a conductive feature layout pattern **426b**.

(118) Conductive feature layout pattern **426a** or conductive feature layout pattern **426b** is coupled to other elements (not shown) of IC structure **100**, **200**. Conductive feature layout pattern **426a** and conductive feature layout pattern **426b** are between first conductive layout pattern **401a** and second conductive layout pattern **401b**. Conductive feature layout pattern **426b** is between power layout pattern **408** and power layout pattern **412**. In some embodiments, conductive feature layout pattern **426a** is between power layout pattern **408** and power layout pattern **412**. In some embodiments, conductive feature layout pattern **426b** is aligned with conductive feature layout pattern **408a** in the second direction Y. Conductive feature layout pattern **426b** has a length $L_{sub.4}$ in the second direction Y. In some embodiments, length $L_{sub.2}$ is the same as length $L_{sub.4}$. In some embodiments, length $L_{sub.2}$ differs from length $L_{sub.4}$.

(119) Conductive feature layout pattern **426b** does not overlap power layout pattern **408** or power layout pattern **412**. Conductive feature layout pattern **426a** has a first color (e.g., color A) and conductive feature layout pattern **426b** has a second color (e.g., color B) different than the first color. In some embodiments, conductive feature layout pattern **426a** and conductive feature layout pattern **426b** have a same color (e.g., color A or B) as each other.

(120) Each of conductive feature layout patterns **402a**, **404a**, **406a**, **408a**, **410a**, **412a**, **420a**, **420b**,

422a, 422b, 424a, 424b, 426a, and 426b is positioned at the M1 layer of layout design **400**. The M1 layer of layout design **400** corresponds to the M1 layer of IC structure **100, 200**. In some embodiments, each of the conductive feature layout patterns of layout design **400** that are located at the M1 layer extends in the same direction. Other configurations of via layers or metal layers are within the scope of the present disclosure.

(121) By power layout pattern **408** being symmetric with respect to second conductive layout pattern **401b**, power layout pattern **404** being symmetric with respect to third conductive layout pattern **401c**, and power layout pattern **412** being symmetric with respect to first conductive layout pattern **401a**, the length $L_{sub.2}$ of conductive feature layout pattern **422b** is less than or equal to distance $D_{sub.4b}$, and the length $L_{sub.4}$ of conductive feature layout pattern **426b** is less than or equal to distance $D_{sub.4a}$. In some embodiments, a minimum value of length L_1 , L_2 , L_3 or L_4 is sufficient to provide corresponding conductive feature layout pattern **420b, 422b, 424b** or **426b** with at least two via access points to provide a connection to upper layers (not shown).

(122) In some embodiments, by not coupling power layout pattern **408** to power layout pattern **412** on the same metal layer (e.g., M1), the space between power layout pattern **408** and power layout pattern **412** can be utilized to form additional routing resources, and the additional routing resources can provide additional via access points to other metal layers. For example, by positioning conductive feature layout patterns **426a** or **426b** between power layout pattern **408** and power layout pattern **412**, conductive feature layout pattern **426a** or **426b** can be utilized to form corresponding conductive structures which provide additional M1 routing resources and via access points to other metal layers, resulting in a layout design **400** that occupies less area than other designs.

(123) In some embodiments, conductive feature layout patterns **420a, 420b, 422a, 422b, 424a, 424b, 426a** or **426b** are utilized to form corresponding conductive structures which provide additional M1 routing resources and via access points capable of being electrically coupled to other elements (not shown) than other designs. By providing additional via access points and additional M1 routing resources, layout design **400** occupies less area than other designs, and IC structure **100, 100', 200, 300** occupies less area than other IC structures.

(124) In some embodiments, one or more of first region **420**, second region **422**, third region **424** or fourth region **426** is a standard cell. In some embodiments, a standard cell is a logic gate cell. In some embodiments, a logic gate cell includes an AND, OR, NAND, NOR, XOR, INV, AND-OR-Invert (AOI), OR-AND-Invert (OAI), MUX, Flip-flop, BUFF, Latch, delay, clock cells, or the like. In some embodiments, a standard cell is a memory cell. In some embodiments, a memory cell includes a static random access memory (SRAM), a dynamic RAM (DRAM), a resistive RAM (RRAM), a magnetoresistive RAM (MRAM) read only memory (ROM), or the like. In some embodiments, a standard cell includes one or more active or passive elements. Examples of active elements include, but are not limited to, transistors and diodes. Examples of transistors include, but are not limited to, metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high voltage transistors, high frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), etc.), FinFETs, planar MOS transistors with raised source/drain, or the like. Examples of passive elements include, but are not limited to, capacitors, inductors, fuses, resistors, or the like. First region **420**, second region **422**, third region **424** or fourth region **426** includes other features not shown for ease of illustration.

(125) Layout design **400** is referred to as a stub-symmetric architecture. For example, power layout patterns **402, 404, 406, 408, 410** and **412** are referred to as stubs; power layout patterns **402, 404** are symmetric with respect to third conductive layout pattern **401c**, power layout patterns **406, 408** are symmetric with respect to second conductive layout pattern **401b**, and power layout patterns **410, 412** are symmetric with respect to first conductive layout pattern **401a**.

(126) FIG. 4B is a diagram of a layout design **400'**, in accordance with some embodiments. Layout

design **400'** is a variation of layout design **400** (FIG. 4A). Components that are the same or similar to those in FIGS. 1A-1B, 2A-2C or 4A are given the same reference numbers, and detailed description thereof is thus omitted. Layout design **400'** is usable to manufacture a portion of an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(127) In comparison with layout design **400** of FIG. 4A, layout design **400'** includes power layout patterns **402'**, **404'**, **406'**, **408'**, **410'** and **412'** instead of corresponding power layout patterns **402**, **404**, **406**, **408**, **410** and **412**, conductive feature layout pattern **422b'** instead of conductive feature layout pattern **422b** (FIG. 4A), and conductive feature layout pattern **424b'** instead of conductive feature layout pattern **424b**.

(128) Power layout patterns **402'**, **404'** are asymmetric with respect to third conductive layout pattern **401c**, power layout patterns **406'**, **408'** are asymmetric with respect to second conductive layout pattern **401b**, and power layout patterns **410'**, **412'** are asymmetric with respect to first conductive layout pattern **401a**. In other words, a center of power layout pattern **402'** or **404'** is not aligned with a center of third conductive layout pattern **401c**. Similarly, a center of power layout pattern **406'** or **408'** is not aligned with a center of second conductive layout pattern **401b**. Similarly, a center of power layout pattern **410'** or **412'** is not aligned with a center of first conductive layout pattern **401a**.

(129) Power layout pattern **402'** or **404'** is separated from corresponding power layout pattern **406'** or **408'** in the second direction by distance $D_{sub.4b'}$. Distance $D_{sub.4b'}$ is less than distance $D_{sub.4b}$ (FIG. 4A).

(130) Power layout pattern **406'** or **408'** is separated from corresponding power layout pattern **410'** or **412'** in the second direction by distance $D_{sub.4a'}$. Distance $D_{sub.4a'}$ is less than distance $D_{sub.4a}$ (FIG. 4A). In some embodiments, distance $D_{sub.4a'}$ or $D_{sub.4b'}$ is based on a height of the first region **420**, second region **422**, third region **424** or fourth region **426**. In some embodiments, a minimum value of distance $D_{sub.4a'}$ or $D_{sub.4b'}$ is sufficient to provide conductive feature layout pattern **420b**, **422b'**, **424b'** or **426b** with at least two via access points to provide a connection to upper layers (not shown).

(131) Conductive feature layout pattern **422b'** has a length $L_{sub.2'}$ and conductive feature layout pattern **424b'** has a length $L_{sub.3'}$.

(132) By the asymmetric orientations of power layout patterns **402'**, **404'**, **406'**, **408'**, **410'** and **412'**, the length $L_{sub.1}$ of conductive feature layout pattern **420b** is less than distance $D_{sub.4b'}$, the length $L_{sub.2'}$ of conductive feature layout pattern **422b'** is less than distance $D_{sub.4b'}$, the length $L_{sub.3'}$ of conductive feature layout pattern **424b'** is less than distance $D_{sub.4a'}$, and the length $L_{sub.4}$ of conductive feature layout pattern **426b** is less than distance $D_{sub.4a'}$ and one of distance $D_{sub.4a'}$ or $D_{sub.4b'}$ is less than one of distance $D_{sub.4a}$ or $D_{sub.4b}$. In some embodiments, a minimum value of length $L_{sub.2'}$ or $L_{sub.3'}$ is sufficient to provide corresponding conductive feature layout patterns **422b'** or **424b'** with at least two via access points to provide a connection to upper layers (not shown).

(133) Layout design **400'** is referred to as a stub-asymmetric architecture. For example, power layout patterns **402'**, **404'**, **406'**, **408'**, **410'** and **412'** are referred to as stubs; power layout patterns **402'**, **404'**, **406'**, **408'**, **410'**, **412'** have asymmetric orientations with respect to first, second and third conductive layout patterns **401a**, **401b**, **401c**.

(134) Power layout patterns **402'**, **404'**, **406'**, **408'**, **410'**, **412'** include corresponding conductive feature layout patterns **402a'**, **404a'**, **406a'**, **408a'**, **410a'**, **412'** instead of corresponding conductive feature layout pattern **402a**, **404a**, **406a**, **408a**, **410a**, **412** (FIG. 4A).

(135) One or more of conductive feature layout pattern **402a'** or **404a'** or cut feature layout pattern **402b**, **402c**, **404b** or **404c** is shifted in the second direction Y relative to first line **440**.

(136) One or more of conductive feature layout pattern **406a'** or **408a'** or cut feature layout pattern **406b**, **406c**, **408b** or **408c** is shifted in the second direction Y relative to second line **442**.

(137) One or more of conductive feature layout pattern **410a'** or **412a'** or cut feature layout pattern

410b, **410c**, **412b** or **412c** is shifted in the second direction Y relative to third line **444**.

(138) In some embodiments, a length of conductive feature layout pattern **402a'**, **404a'**, **406a'**, **408a'**, **410a'** or **412a'** is greater than a length of conductive feature layout pattern **402a**, **404a**, **406a**, **408a**, **410a** or **412a**.

(139) In some embodiments, conductive feature layout patterns **420a**, **420b**, **422a**, **422b'**, **424a**, **424b'**, **426a** or **426b** are utilized to form corresponding conductive structures which provide additional M1 routing resources and via access points capable of being electrically coupled to other elements (not shown) than other designs. By providing additional via access points and additional M1 routing resources, layout design **400'** occupies less area than other designs, and IC structure **100**, **100'**, **200**, **300** occupies less area than other IC structures.

(140) FIG. 5A is a diagram of a layout design **500**, in accordance with some embodiments. Layout design **500** is usable to manufacture a portion of IC structure **300** (FIGS. 3A-3C). Components that are the same or similar to those in FIGS. 1A-1B, 2A-2C, 3A-3C or 4A-4B are given the same reference numbers, and detailed description thereof is thus omitted.

(141) Layout design **500** includes portions of layout design **400**, a conductive feature layout pattern **522** and a conductive feature layout pattern **526**. In comparison with layout design **400** of FIG. 4A, layout design **500** does not include power layout patterns **402**, **408** and **410**. By removing power layout patterns **402**, **408** and **410**, layout design **500** provides additional M1 routing resources than other designs.

(142) In some embodiments, by removing power layout patterns **402** or **410**, the length L.sub.1 of conductive feature layout pattern **420b** or the length L.sub.3 of conductive feature layout pattern **424b** is increased to another length provided conductive feature layout pattern **420b** and conductive feature layout pattern **424b** do not overlap each other.

(143) In some embodiments, by removing power layout pattern **408**, at least the length L.sub.2 of conductive feature layout pattern **422b** or the length L.sub.4 of conductive feature layout pattern **426b** is increased to another length provided conductive feature layout pattern **422b** and conductive feature layout pattern **426b** do not overlap each other.

(144) Power layout pattern **406** is separated from power layout pattern **404** in the second direction by distance D.sub.4b. Power layout pattern **406** is separated from power layout pattern **412** in the second direction by distance D.sub.4a.

(145) In some embodiments, the length L.sub.1 of conductive feature layout pattern **420b** of layout design **500** is greater than distance D.sub.4b provided conductive feature layout pattern **420b** does not overlap other layout features positioned at the M1 layer of layout design **500**. The M1 layer of layout design **500** corresponds to the M1 layer of IC structure **100**, **100'**, **200**, **300**. In some embodiments, the length L.sub.2 of conductive feature layout pattern **422b** of layout design **500** is greater than distance D.sub.4b provided conductive feature layout pattern **422b** does not overlap other layout features positioned at the M1 layer of layout design **500**. Other configurations of via layers or metal layers are within the scope of the present disclosure.

(146) In some embodiments, the length L.sub.3 of conductive feature layout pattern **424b** of layout design **500** is greater than distance D.sub.4a provided conductive feature layout pattern **424b** does not overlap other layout features positioned at the M1 layer of layout design **500**. In some embodiments, the length L.sub.4 of conductive feature layout pattern **426b** of layout design **500** is greater than distance D.sub.4a provided conductive feature layout pattern **426b** does not overlap other layout features positioned at the M1 layer of layout design **500**. Other configurations of via layers or metal layers are within the scope of the present disclosure.

(147) Conductive feature layout pattern **522** has a length L.sub.5. Conductive feature layout pattern **522** includes conductive feature layout pattern **422b** (FIG. 4A) and conductive feature layout pattern **522b**. Conductive feature layout pattern **522b** corresponds to an extended portion of conductive feature layout pattern **422b** in the second direction Y. Conductive feature layout pattern **522b** has a length L.sub.2a in the second direction Y. By increasing the length of conductive feature

layout pattern **422b** to length $L_{sub.5}$, conductive feature layout pattern **522** has additional routing resources than other approaches.

(148) Conductive feature layout pattern **526** has a length $L_{sub.5'}$. Conductive feature layout pattern **526** includes conductive feature layout pattern **426b** (FIG. 4A) and conductive feature layout pattern **526b**. Conductive feature layout pattern **526b** corresponds to an extended portion of conductive feature layout pattern **426b** in the second direction Y. Conductive feature layout pattern **526b** has a length $L_{sub.4a}$ in the second direction Y. By increasing the length of conductive feature layout pattern **426b** to length $L_{sub.5'}$, conductive feature layout pattern **522** has additional routing resources than other approaches. In some embodiments, a minimum value of length L_5 or $L_{5'}$ is sufficient to provide corresponding conductive feature layout patterns **522** or **526** with at least two via access points to provide a connection to upper layers (not shown).

(149) Conductive feature layout pattern **522** and **526** do not overlap each other. Conductive feature layout pattern **522** and conductive feature layout pattern **526** are positioned at the M1 layer of layout design **500**. Other configurations of via layers or metal layers are within the scope of the present disclosure.

(150) Layout design **500** is referred to as an alternating stub-symmetric architecture. For example, power layout patterns **404**, **406** and **412** are arranged as alternating stubs in the first direction X; power layout patterns **404**, **406** and **412** have asymmetric orientations with respect to first, second and third conductive layout patterns **401a**, **401b**, **401c**.

(151) In some embodiments, conductive feature layout patterns **420a**, **420b**, **422a**, **522**, **424a**, **424b**, **426a** or **526** are utilized to form corresponding conductive structures which provide additional M1 routing resources and via access points capable of being electrically coupled to other elements (not shown) than other designs. By providing additional via access points and additional M1 routing resources, layout design **500** occupies less area than other designs, and IC structure **100**, **100'**, **200**, **300** occupies less area than other IC structures.

(152) FIG. 5B is a diagram of a layout design **500'**, in accordance with some embodiments. Layout design **500'** is a variation of layout design **400** (FIG. 4A) or layout design **500** (FIG. 5A).

Components that are the same or similar to those in FIGS. 1A-1B, 2A-2C, 3A-3C, 4A-4B or 5A are given the same reference numbers, and detailed description thereof is thus omitted. Layout design **500'** is usable to manufacture a portion of an IC structure similar to IC structure **300** (FIGS. 3A-3C).

(153) Layout design **500'** includes portions of layout design **400'**, a conductive feature layout pattern **522'** and a conductive feature layout pattern **526'**. In comparison with layout design **400'** of FIG. 4B, layout design **500'** does not include power layout patterns **402'**, **408'** and **410'**, and conductive feature layout pattern **426b'** replaces conductive feature layout pattern **426b**. By removing power layout patterns **402'**, **408'** and **410'**, layout design **500'** provides additional M1 routing resources than other designs.

(154) In some embodiments, by removing power layout patterns **402'** and **410'**, the length $L_{sub.1}$ of conductive feature layout pattern **420b** or the length $L_{sub.3}$ of conductive feature layout pattern **424b'** is increased to another length provided conductive feature layout pattern **420b** and conductive feature layout pattern **424b'** do not overlap each other.

(155) In some embodiments, by removing power layout patterns **408'**, the length $L_{sub.2'}$ of conductive feature layout pattern **422b'** or the length $L_{sub.4'}$ of conductive feature layout pattern **426b'** is increased to another length (e.g., $L_{sub.6}$ or $L_{sub.6'}$) provided conductive feature layout pattern **422b'** and conductive feature layout pattern **426b'** do not overlap each other.

(156) Power layout pattern **406'** is separated from power layout pattern **404'** in the second direction by distance $D_{sub.4b'}$. Power layout pattern **406'** is separated from power layout pattern **412'** in the second direction by distance $D_{sub.4a'}$.

(157) In some embodiments, if the length $L_{sub.1}$ of conductive feature layout pattern **420b** of layout design **500'** is greater than distance $D_{sub.4b'}$, then the length $L_{sub.3'}$ of conductive feature

layout pattern **424b'** of layout design **500'** is less than distance D.sub.4a' and length L.sub.1.

(158) In some embodiments, if the length L.sub.3' of conductive feature layout pattern **424b'** of layout design **500'** is greater than distance D.sub.4a', then length L.sub.1 of conductive feature layout pattern **420b** of layout design **500'** is less than distance D.sub.4b' and length L.sub.3'.

(159) Conductive feature layout pattern **522'** has a length L.sub.6 in the second direction Y. Conductive feature layout pattern **522'** includes conductive feature layout pattern **422b'** (FIG. 4B) and a conductive feature layout pattern **522b'**. Conductive feature layout pattern **522b'** corresponds to an extended portion of conductive feature layout pattern **422b'** in the second direction Y. Conductive feature layout pattern **522b'** has a length L.sub.2b in the second direction Y. By increasing the length L.sub.2' of conductive feature layout pattern **422b'** to length L.sub.6, conductive feature layout pattern **522'** has additional routing resources than other approaches.

(160) Conductive feature layout pattern **526'** has a length L.sub.6' in the second direction Y. Conductive feature layout pattern **526'** includes conductive feature layout pattern **426b'** (FIG. 4B) and a conductive feature layout pattern **526b'**. Conductive feature layout pattern **526b'** corresponds to an extended portion of conductive feature layout pattern **426b'** in the second direction Y. Conductive feature layout pattern **526b'** has a length Lab in the second direction Y. By increasing the length L.sub.4' of conductive feature layout pattern **426b'** to length L.sub.6', conductive feature layout pattern **526'** has additional routing resources than other approaches.

(161) In some embodiments, if the length L.sub.6 of conductive feature layout pattern **522'** of layout design **500'** is greater than distance D.sub.4b', then the length L.sub.6' of conductive feature layout pattern **526'** of layout design **500'** is less than distance D.sub.4a' and length L.sub.6.

(162) In some embodiments, if the length L.sub.6' of conductive feature layout pattern **526'** of layout design **500'** is greater than distance D.sub.4a', then length L.sub.6 of conductive feature layout pattern **522'** of layout design **500'** is less than distance D.sub.4b' and length L.sub.6'. In some embodiments, a minimum value of length L6 or L6' is sufficient to provide corresponding conductive feature layout patterns **522'** or **526'** with at least two via access points to provide a connection to upper layers (not shown).

(163) Conductive feature layout pattern **522'** and conductive feature layout pattern **526'** do not overlap each other. Conductive feature layout pattern **522'** and conductive feature layout pattern **526'** are positioned at the M1 layer of layout design **500'**, and do not overlap other layout features positioned at the M1 layer of layout design **500'**. Other configurations of via layers or metal layers are within the scope of the present disclosure.

(164) Layout design **500'** is referred to as an alternating stub-asymmetric architecture. For example, power layout patterns **404'**, **406'** and **412'** are arranged as alternating stubs in the first direction X; power layout patterns **404'**, **406'** and **412'** have asymmetric orientations with respect to first, second and third conductive layout patterns **401a**, **401b**, **401c**.

(165) In some embodiments, conductive feature layout patterns **420a**, **420b**, **422a**, **522**, **424a**, **424b'**, **426a** or **526'** are utilized to form corresponding conductive structures which provide additional M1 routing resources and via access points capable of being electrically coupled to other elements (not shown) than other designs. By providing additional via access points and additional M1 routing resources, layout design **500'** occupies less area than other designs, and IC structure **100**, **100'**, **200**, **300** occupies less area than other IC structures.

(166) FIG. 6 is a diagram of a layout design **600**, in accordance with some embodiments. Layout design **600** is a variation of layout design **400**. Components that are the same or similar to those in FIGS. 1A-1B, 2A-2C, 3A-3C, 4A-4B, 5A-5 and 6-18 (shown below) are given the same reference numbers, and detailed description thereof is thus omitted. Layout design **600** is usable to manufacture a portion of an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(167) Layout design **600** includes features of first region **420** and second region **422** of layout design **400** of FIG. 4A. For example, from layout design **400** of FIG. 4A, layout design **600** includes second conductive layout pattern **401b**, third conductive layout pattern **401c**, power layout

patterns **402**, **404**, **406**, **408**, first region **420** and second region **422**.

(168) In comparison with layout design **400** of FIG. **4A**, layout design **600** further includes conductive feature layout pattern **208b'**, conductive feature layout pattern **208c'**, via layout patterns **204a'**, **204b'**, **206a'**, **206b'**, **212a'**, **214a'**, conductive feature layout patterns **220a'**, **220b'**, conductive feature layout patterns **602a**, **602b**, . . . , **602i**, and conductive feature layout patterns **604a**, **604b**, . . . , **604d**.

(169) Conductive feature layout pattern **208b'** and conductive feature layout pattern **208c'** are usable to manufacture second conductive structure **208b** and third conductive structure **208c**, respectively.

(170) Via layout patterns **204a'**, **204b'**, **206a'**, **206b'**, **212a'**, **214a'** are usable to manufacture vias **204a**, **204b**, **206a**, **206b**, **212a**, **214a**, respectively.

(171) Conductive feature layout patterns **220a'**, **220b'** are usable to manufacture conductive structures **220a**, **220b**, respectively.

(172) Conductive feature layout pattern **602a** is a variation of conductive feature layout pattern **420b**, and conductive feature layout pattern **602i** is a variation of conductive feature layout pattern **422b**. Conductive feature layout pattern, **602b**, . . . , **602h** are variations of conductive feature layout patterns **420a** or **420b**.

(173) Each of conductive feature layout patterns **602a**, **602b**, . . . , **602i** extend in the second direction Y, and are separated from each other in the first direction X. Each of conductive feature layout patterns **602a**, **602b**, . . . , **602i** are aligned with a corresponding gridline **610a**, **610b**, . . . , **610i**. A distance between all adjacent conductive feature layout patterns **602a**, **602b**, . . . , **602i** is uniform. In some embodiments, a distance between a pair of adjacent conductive feature layout patterns **602a**, **602b**, . . . , **602i** differs from a distance between another pair of adjacent conductive feature layout patterns **602a**, **602b**, . . . , **602i**. Conductive feature layout patterns **602a**, **602b**, . . . , **602i** are positioned at the M1 layer of layout design **600**. Other configurations of via layout patterns and corresponding via layers or conductive feature layout patterns and corresponding metal layers are within the scope of the present disclosure. Conductive feature layout patterns **602a**, **602b**, . . . , **602i** provide access to other underlying or overlying layers in IC structure **100**, **200**. In some embodiments, conductive feature layout patterns **602a**, **602b**, . . . , **602i** are coupled to via layout patterns (not shown) that couple conductive feature layout patterns **602a**, **602b**, . . . , **602i** to other portions of layout design **600**.

(174) Conductive feature layout pattern **604a**, **604b**, . . . , **604d** extend in the first direction X, and are separated from each other in the second direction Y. A distance between all adjacent conductive feature layout patterns **604a**, **604b**, . . . , **604d** is uniform. In some embodiments, a distance between a pair of adjacent conductive feature layout patterns **604a**, **604b**, . . . , **604d** differs from a distance between another pair of adjacent conductive feature layout patterns **604a**, **604b**, . . . , **604d**. Conductive feature layout patterns **604a**, **604b**, . . . , **604d** are positioned at the M0 layer of layout design **600**. Other configurations of via layout patterns and corresponding via layers or conductive feature layout patterns and corresponding metal layers are within the scope of the present disclosure.

(175) Conductive feature layout patterns **602a**, **602b**, . . . , **602i** or conductive feature layout patterns **604a**, **604b**, . . . , **604i** are coupled to other elements (not shown). In some embodiments, a length of a conductive feature layout pattern of conductive feature layout patterns **602a** or **602i** of FIG. **6** is less than a length of one or more conductive feature layout patterns of conductive feature layout patterns **602b**, . . . , or **602h**.

(176) An M1 stub pitch **P4** is the center to center pitch between adjacent power layout patterns of a same row in the second direction X. For example, pitch **P4** is the center to center pitch between power layout patterns **402**, **406** and corresponding power layout patterns **404**, **408**.

(177) An M3 strap pitch **P4'** is the center to center pitch between adjacent M3 straps. For example, pitch **P4'** is the center to center pitch between conductive feature layout pattern **220a'** and

conductive feature layout pattern **220b'**. In some embodiments, the M1 stub pitch **P4** or the M3 strap pitch **P4'** is measured from gridline **610a** to gridline **610i**. In some embodiments, the M1 stub pitch **P4** is equal to the M3 strap pitch **P4'**.

(178) In some embodiments, conductive feature layout patterns **602a**, **602b**, . . . , **602i** are utilized to form corresponding conductive structures which provide additional M1 routing resources and via access points capable of being electrically coupled to other elements (not shown) than other designs. By providing additional via access points and additional M1 routing resources, layout design **600** occupies less area than other designs, and IC structure **100**, **100'**, **200**, **300** occupies less area than other IC structures.

(179) FIG. 7 is a diagram of a layout design **700**, in accordance with some embodiments. Layout design **700** is another variation of layout design **400** or **400'**. Layout design **700** is usable to manufacture an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(180) In comparison with layout design **600** of FIG. 6, power layout patterns **404**, **408** and conductive feature layout pattern **220b'** of layout design **700** are shifted from being positioned over gridline **610i** to being positioned over gridline **610e**, and layout design **700** further includes power layout patterns **702**, **704**, conductive feature layout pattern **220e'** and via layout patterns **204e'**, **206e'**, **214c'**.

(181) By including additional power layout patterns (e.g., power layout patterns **702**, **704**), a pitch **P5** between adjacent power layout patterns of the same row is reduced and results in additional M1 routing resources or additional via access points than other designs.

(182) Power layout patterns **702**, **704**, conductive feature layout pattern **220e'** and via layout patterns **204e'**, **206e'**, **214c'** are positioned over gridline **610i**.

(183) Via layout patterns **204e'**, **206e'**, **214c'** are usable to manufacture vias **204e**, **206e**, **214c**, respectively.

(184) Power layout pattern **702** is usable to manufacture conductive structure **114e** and via **112e** (FIGS. 1A-1B, 2A-2C). Power layout pattern **704** is usable to manufacture conductive structure **110e** and via **108e** (FIGS. 1A-1B, 2A-2C).

(185) Power layout pattern **702** is similar to power layout pattern **402** with the reference numerals increased by 300. Conductive feature layout pattern **702a** is usable to manufacture conductive structure **114e** (FIGS. 1A-1B, 2A-2C). Cut feature layout patterns **702b**, **702c** is similar to cut feature layout patterns **402b**, **402c**, respectively. In some embodiments, the cut feature layout pattern **702b** or **702c** identifies a location of the conductive structure **114e** that is removed during fabrication of IC structure **100**, **100'**. Via layout pattern **702d** is usable to manufacture via **112e** (FIGS. 1A-1B, 2A-2C).

(186) Power layout pattern **704** is similar to power layout pattern **404** with the reference numerals increased by 300. Conductive feature layout pattern **704a** is usable to manufacture conductive structure **110e** (FIGS. 1A-1B, 2A-2C). Cut feature layout patterns **704b**, **704c** is similar to cut feature layout patterns **406b**, **406c**, respectively. In some embodiments, the cut feature layout pattern **704b** or **704c** identifies a location of the conductive structure **110e** that is removed during fabrication of IC structure **100**, **100'**. Via layout pattern **704d** is usable to manufacture via **108e** (FIGS. 1A-1B, 2A-2C).

(187) Conductive feature layout pattern **220e'** is usable to manufacture conductive structure **220e**.

(188) In some embodiments, a length of a conductive feature layout pattern of conductive feature layout patterns **602a**, **602e** or **602i** of FIG. 7 is less than one or more conductive feature layout patterns of conductive feature layout patterns **602b**, **602c**, **602d**, **602f**, **602g** or **602h**.

(189) An M1 stub pitch **P5** is the center to center pitch between adjacent power layout patterns of a same row in the second direction **X**. For example, pitch **P5** is the center to center pitch between power layout patterns **402**, **406** and corresponding power layout patterns **404**, **408**. Similarly, **P5** is the center to center pitch between power layout patterns **404**, **408** and corresponding power layout patterns **702**, **704**. Other configurations of via layout patterns and corresponding via layers or

conductive feature layout patterns and corresponding metal layers are within the scope of the present disclosure.

(190) An M3 strap pitch P5' is the center to center pitch between adjacent M3 straps. For example, pitch P5' is the center to center pitch between either (1) conductive feature layout pattern **220a'** and conductive feature layout pattern **220b'**, or (2) conductive feature layout pattern **220b'** and conductive feature layout pattern **220e'**.

(191) In some embodiments, the M1 stub pitch P5 or the M3 strap pitch P5' is measured from gridline **610a** to gridline **610e**, or from gridline **610e** to gridline **610i**. In some embodiments, the M1 stub pitch P5 is equal to the M3 strap pitch P5'. In some embodiments, the M1 stub pitch P5 differs from the M3 strap pitch P5'.

(192) In some embodiments, conductive feature layout patterns **602a**, **602b**, . . . , **602i** are utilized to form corresponding conductive structures which provide additional M1 routing resources and via access points capable of being electrically coupled to other elements (not shown) than other designs. By providing additional via access points and additional M1 routing resources, layout design **700** occupies less area than other designs, and IC structure **100**, **100'**, **200**, **300** occupies less area than other IC structures.

(193) FIG. **8** is a diagram of a layout design **800**, in accordance with some embodiments. Layout design **800** is another variation of layout design **400** or **400'**. Layout design **800** is usable to manufacture an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(194) In comparison with layout design **700** of FIG. 7, power layout patterns **404**, **408** and conductive feature layout pattern **220b'** of layout design **800** are shifted from being positioned over gridline **610e** to being positioned over gridline **610c**, and layout design **800** further includes power layout patterns **802**, **804**, **806**, **808**, conductive feature layout patterns **220c'**, **220d'** and via layout patterns **204c'**, **204d'**, **206c'**, **206d'**, **212b'**, **214b'**.

(195) By including additional power layout patterns (e.g., power layout patterns **802**, **804**, **806**, **808**), a pitch P6 between adjacent power layout patterns of the same row is reduced and results in additional M1 routing resources or additional via access points than other designs.

(196) Power layout patterns **802**, **806**, conductive feature layout pattern **220c'** and via layout patterns **204c'**, **206c'**, **214b'** are positioned over gridline **610e**.

(197) Power layout patterns **804**, **808**, conductive feature layout pattern **220d'** and via layout patterns **204d'**, **206d'**, **212b'** are positioned over gridline **610g**.

(198) Via layout patterns **204c'**, **204d'**, **206c'**, **206d'**, **212b'**, **214b'** are usable to manufacture vias **204c**, **204d**, **206c**, **206d**, **212b**, **214b**, respectively.

(199) Power layout patterns **802**, **804**, **806**, **808** are corresponding power layout patterns **402**, **404**, **406**, **408** with the reference numerals increased by 400.

(200) Power layout pattern **802** is usable to manufacture conductive structure **114c** and via **112c** (FIGS. 1A-1B, 2A-2C). Power layout pattern **804** is usable to manufacture conductive structure **114d** and via **112d** (FIGS. 1A-1B, 2A-2C). Power layout pattern **806** is usable to manufacture conductive structure **110c** and via **108c** (FIGS. 1A-1B, 2A-2C). Power layout pattern **808** is usable to manufacture conductive structure **110d** and via **108d** (FIGS. 1A-1B, 2A-2C).

(201) Conductive feature layout patterns **802a**, **804a**, **806a**, **808a** are usable to manufacture corresponding conductive structures **114c**, **114d**, **110c**, **110d** (FIGS. 1A-1B, 2A-2C).

(202) Via layout patterns **802d**, **804d**, **806d**, **808d** are usable to manufacture corresponding vias **112c**, **112d**, **108c**, **108d** (FIGS. 1A-1B, 2A-2C).

(203) Cut feature layout patterns **802b**, **802c**, **804b**, **804c**, **806b**, **806c**, **808b**, **808c** are corresponding cut feature layout patterns **402b**, **402c**, **404b**, **404c**, **406b**, **406c**, **408b**, **408c** with the reference numerals increased by 400. In some embodiments, the cut feature layout pattern **802b** or **802c** identifies a location of the conductive structure **114c** that is removed during fabrication of IC structure **100**, **100'**, **200** or **300**. In some embodiments, the cut feature layout pattern **804b** or **804c** identifies a location of the conductive structure **114d** that is removed during fabrication of IC

structure **100**, **100'**, **200** or **300**. In some embodiments, the cut feature layout pattern **806b** or **806c** identifies a location of the conductive structure **110c** that is removed during fabrication of IC structure **100**, **100'**, **200** or **300**. In some embodiments, the cut feature layout pattern **808b** or **808c** identifies a location of the conductive structure **110d** that is removed during fabrication of IC structure **100**, **100'**, **200** or **300**.

(204) Conductive feature layout patterns **220c'**, **220d'** are usable to manufacture corresponding conductive structures **220c**, **220d**.

(205) In some embodiments, a length of a conductive feature layout pattern of conductive feature layout patterns **602a**, **602c**, **602e**, **602g** or **602i** of FIG. 7 is less than one or more conductive feature layout patterns of conductive feature layout patterns **602b**, **602d**, **602f** or **602h**.

(206) An M1 stub pitch **P6** is the center to center pitch between adjacent power layout patterns of a same row in the second direction X. For example, pitch **P6** is the center to center pitch between power layout pattern **402**, **406** and corresponding power layout patterns **404**, **408**. Similarly, **P6** is the center to center pitch between power layout patterns **404**, **408** and corresponding power layout patterns **802**, **806**. Other configurations of via layout patterns and corresponding via layers or conductive feature layout patterns and corresponding metal layers are within the scope of the present disclosure.

(207) An M3 strap pitch **P6'** is the center to center pitch between adjacent M3 straps. For example, pitch **P6'** is the center to center pitch between conductive feature layout pattern **220a'** and conductive feature layout pattern **220b'**. Similarly, pitch **P6'** is the center to center pitch between conductive feature layout pattern **220b'** and conductive feature layout pattern **220c'**.

(208) In some embodiments, the M1 stub pitch **P6** or the M3 strap pitch **P6'** is measured from: gridline **610a** to gridline **610c**, gridline **610c** to gridline **610e**, gridline **610e** to gridline **610g**, or gridline **610g** to gridline **610i**. In some embodiments, the M1 stub pitch **P6** is equal to the M3 strap pitch **P6'**. In some embodiments, the M1 stub pitch **P6** differs from the M3 strap pitch **P6'**.

(209) FIG. 9 is a diagram of a layout design **900**, in accordance with some embodiments. Layout design **900** is a variation of layout design **400** or **400'**. Layout design **900** is usable to manufacture an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(210) In comparison with layout design **600** of FIG. 6, conductive feature layout pattern **208c'** and via layout patterns **206a'**, **206b'**, **214a'**, **402d**, **404d** of layout design **900** are shifted in the second direction Y from first line **440** to a boundary **910a** by an offset distance **D.sub.6'**, and conductive feature layout pattern **208b'** and via layout patterns **204a'**, **204b'**, **212a'**, **406d**, **408d** of layout design **900** are shifted in the second direction Y from second line **442** to a boundary **910b** by an offset distance **D.sub.6**.

(211) By offsetting conductive feature layout pattern **208c'**, **208b'** by corresponding offset distances **D.sub.6'**, **D.sub.6**, and second and third conductive feature layout patterns **401b**, **401c** are exposed to form corresponding conductive structures that provide additional M0 via access points and additional M1 routing resources than other designs.

(212) A center of conductive feature layout pattern **208c'** and a center of via layout patterns **206a'**, **206b'**, **214a'**, **402d**, **404d** of layout design **900** are aligned with boundary **910a**. In some embodiments, a center of conductive feature layout pattern **208c'** and a center of each of via layout patterns **206a'**, **206b'**, **214a'**, **402d**, **404d** is aligned with one another in the first direction X.

(213) A center of conductive feature layout pattern **208b'** and a center of via layout patterns **204a'**, **204b'**, **212a'**, **406d**, **408d** of layout design **900** are aligned with boundary **910b**. In some embodiments, a center of conductive feature layout pattern **208b'** and a center of each of via layout patterns **204a'**, **204b'**, **212a'**, **406d**, **408d** is aligned with one another in the first direction X.

(214) In some embodiments, offset distance **D.sub.6'** is equal to offset distance **D.sub.6**. In some embodiments, offset distance **D.sub.6'** is not equal to offset distance **D.sub.6**.

(215) FIG. 10 is a diagram of a layout design **1000**, in accordance with some embodiments. Layout design **1000** is another variation of layout design **400** or **400'**. Layout design **1000** is usable to

manufacture an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(216) In comparison with layout design **900** of FIG. 9, via layout patterns **402d**, **404d** of layout design **1000** are shifted in the second direction Y from boundary **910a** to first line **440** by offset distance $D_{\text{sub.6}}$, and via layout patterns **406d**, **408d** of layout design **1000** are shifted in the second direction Y from boundary **910b** to second line **442** by offset distance $D_{\text{sub.6}}$.

(217) By offsetting via layout patterns **402d**, **404d**, **406d**, **408d** by offset distances $D_{\text{sub.6}}$, $D_{\text{sub.6}}$, conductive feature layout patterns **402a**, **404a**, **406a**, **408a** are exposed to form corresponding conductive structures that provide additional M1 via access points and additional M1 routing resources than other designs.

(218) A center of via layout patterns **402d**, **404d** of layout design **1000** is aligned with first line **440**. A center of conductive feature layout pattern **208c'** and a center of via layout patterns **206a'**, **206b'**, **214a'** of layout design **1000** are aligned with boundary **910a**. In some embodiments, a center of conductive feature layout pattern **208c'** and a center of each of via layout patterns **206a'**, **206b'**, **214a'** is aligned with one another in the first direction X.

(219) A center of via layout patterns **406d**, **408d** of layout design **1000** are aligned with second line **442**. A center of conductive feature layout pattern **208b'** and a center of via layout patterns **204a'**, **204b'**, **212a'** of layout design **1000** are aligned with boundary **910b**. In some embodiments, a center of conductive feature layout pattern **208b'** and a center of each of via layout patterns **204a'**, **204b'**, **212a'** is aligned with one another in the first direction X.

(220) FIG. 11 is a diagram of a layout design **1100**, in accordance with some embodiments. Layout design **1100** is another variation of layout design **400** or **400'**. Layout design **1100** is usable to manufacture an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(221) In comparison with layout design **900** of FIG. 9, layout design **1100** includes power layout patterns **402'**, **404'**, **406'** and **408'** instead of corresponding power layout patterns **402**, **404**, **406** and **408**.

(222) Power layout patterns **402'**, **404'** are asymmetric with respect to third conductive layout pattern **401c**. For example, in comparison with layout design **900** of FIG. 9, power layout patterns **402'** and **404'** of layout pattern **1100** are shifted in the second direction Y from corresponding power layout patterns **402** and **404** by an offset distance $D_{\text{sub.6a}}$ such that a center of power layout patterns **402'** or **404'** is aligned with a center of third conductive layout pattern **401c** along first line **440**. First line **440** is separated from a boundary **1110a** by offset distance $D_{\text{sub.6a}}$. While power layout patterns **402'**, **404'** are shifted in the second direction Y with respect to boundary **1110a**, a center of conductive feature layout pattern **208c'** and a center of via layout patterns **206a'**, **206b'**, **214a'**, **402d**, **404d** of layout design **1100** are aligned with boundary **910a**.

(223) Power layout patterns **406'**, **408'** are asymmetric with respect to second conductive layout pattern **401b**. For example, in comparison with layout design **900** of FIG. 9, power layout patterns **406'** and **408'** of layout pattern **1100** are shifted in the second direction Y from corresponding power layout patterns **406** and **408** by an offset distance $D_{\text{sub.6b}}$ such that a center of power layout patterns **406'** or **408'** is aligned with a center of second conductive layout pattern **401b** along second line **442**. Second line **442** is separated from a boundary **1110b** by offset distance $D_{\text{sub.6b}}$. While power layout patterns **406'**, **408'** are shifted in the second direction Y with respect to boundary **1110b**, a center of conductive feature layout pattern **208b'** and a center of via layout patterns **204a'**, **204b'**, **212a'**, **406d**, **408d** of layout design **1100** are aligned with boundary **910b**.

(224) By offsetting power layout patterns **402'**, **404'**, **406'**, **408'** by corresponding offset distances $D_{\text{sub.6a}}$, $D_{\text{sub.6b}}$, second and third conductive feature layout patterns **401b**, **401c** are exposed to form corresponding conductive structures that provide additional M0 via access points, and conductive feature layout patterns **402a'**, **404a'**, **406a'**, **408a'** are exposed to form corresponding conductive structures that provide additional M1 via access points and additional M1 routing resources than other designs.

(225) In some embodiments, one or more distances of offset distance $D_{\text{sub.6a}}$ or $D_{\text{sub.6b}}$ is equal

to a distance of distance D.sub.6' or D.sub.6. In some embodiments, one or more distances of offset distance D.sub.6a or D.sub.6b differs from a distance of distance D.sub.6' or D.sub.6.

(226) In some embodiments, a length of conductive feature layout patterns **602a** or **602i** of FIG. **11** is less than one or more of conductive feature layout patterns **602b**, **602c**, . . . , or **602h**. In some embodiments, a length of conductive feature layout patterns **602a** or **602i** of layout pattern **1100** is less than one or more of conductive feature layout patterns **602a** or **602i** of layout pattern **900**.

(227) FIG. **12** is a diagram of a layout design **1200**, in accordance with some embodiments. Layout design **1200** is another variation of layout design **400** or **400'**. Layout design **1200** is usable to manufacture an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(228) In comparison with layout design **1000** of FIG. **10**, layout design **1200** includes: a second conductive layout pattern **1201a** instead of second conductive layout pattern **401b**, a third conductive layout pattern **1201b** instead of third conductive layout pattern **401c** and conductive feature layout patterns **1202a**, **1202b**, . . . , **1202e** instead of conductive feature layout patterns **604a**, **604b**, . . . , **604d**.

(229) Conductive feature layout patterns **1202a**, **1202b**, . . . , **1202e** are conductive feature layout patterns **604a**, **604b**, . . . , **604d**.

(230) Second conductive layout pattern **1201a** is a variation of second conductive layout pattern **401b**, and is usable to manufacture second rail **102b** (FIGS. 1A-1B, 2A-2C, 3A-3C). Second conductive layout pattern **1201a** has a width W.sub.2a in the second direction Y.

(231) Third conductive layout pattern **1201b** is a variation of third conductive layout pattern **401c**, and is usable to manufacture third rail **102c** (FIGS. 1A-1B, 2A-2C, 3A-3C). Third conductive layout pattern **1201b** has a width W.sub.2b in the second direction Y.

(232) By offsetting conductive feature layout pattern **208c'** by offset distance D.sub.6', and increasing the widths W.sub.2a, W.sub.2b of corresponding second and third conductive feature layout patterns **1201b**, **1201c**, conductive feature layout patterns **402a**, **404a**, **406a**, **408a**, and second and third conductive feature layout patterns **1201b**, **1201c** are further exposed to form corresponding conductive structures that provide additional via access points, and additional M1 routing resources than other designs.

(233) In some embodiments, width W.sub.2a is the same as width W.sub.2b. In some embodiments, width W.sub.2a differs from width W.sub.2b. In some embodiments, at least one width of widths W.sub.1a or W.sub.1b of layout pattern **1200** is less than at least one width of width W.sub.2a or W.sub.2b.

(234) In FIG. **12**, the center of conductive feature layout pattern **208b'** and the center of via layout patterns **204a'**, **204b'**, **212a'**, **406d** and **408d** are aligned with the second line **442**. By using conductive feature layout patterns **1202a**, **1202b**, . . . , **1202e**, second conductive layout pattern **1201a** and third conductive layout pattern **1201b**, a height of layout design **1200** is greater than a height of layout design **1000**.

(235) FIG. **13** is a diagram of a layout design **1300**, in accordance with some embodiments. Layout design **1300** is another variation of layout design **400** or **400'**. Layout design **1300** is usable to manufacture an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(236) In comparison with layout design **1200** of FIG. **12**, via layout patterns **402d**, **404d** of layout design **1300** are shifted in the second direction Y from first line **440** to boundary **910a** by an offset distance D.sub.6', via layout patterns **406d**, **408d** of layout design **1300** are shifted in the second direction Y from second line **442** to boundary **910b** by an offset distance D.sub.6, and conductive feature layout pattern **208b'** and via layout patterns **204a'**, **204b'**, **212a'** of layout design **1300** are shifted in the second direction Y from second line **442** to a boundary **1310b** by an offset distance D.sub.7. Boundary **1310b** is separated from second line **442** in the second direction Y by distance D.sub.7.

(237) By offsetting conductive feature layout pattern **208b'**, **208c'** by corresponding offset distances D.sub.7, D.sub.6' and increasing the widths W.sub.2a, W.sub.2b of corresponding second and third

conductive feature layout patterns **1201b**, **1201c**, conductive feature layout patterns **402a**, **404a**, **406a**, **408a**, and second and third conductive feature layout patterns **1201b**, **1201c** are further exposed to form corresponding conductive structures that provide additional via access points, and additional M1 routing resources than other designs.

(238) A center of conductive feature layout pattern **208c'** and a center of via layout patterns **206a'**, **206b'**, **214a'**, **402d**, **404d** of layout design **1300** are aligned with boundary **910a**. A center of third conductive layout pattern **1201b** is aligned with first line **440**. In some embodiments, a center of conductive feature layout pattern **208c'** and a center of each of via layout patterns **206a'**, **206b'**, **214a'**, **402d**, **404d** is aligned with one another in the first direction X.

(239) A center of via layout patterns **406d**, **408d** of layout design **1300** is aligned with boundary **910b**. A center of second conductive layout pattern **1201a** is aligned with second line **442**.

(240) A center of conductive feature layout pattern **208b'** and a center of via layout patterns **204a'**, **204b'**, **212a'** of layout design **1300** are aligned with boundary **1310b**. In some embodiments, a center of conductive feature layout pattern **208b'** and a center of each of via layout patterns **204a'**, **204b'**, **212a'** is aligned with one another in the first direction X.

(241) FIG. **14** is a diagram of a layout design **1400**, in accordance with some embodiments. Layout design **1400** is another variation of layout design **400** or **400'**. Layout design **1400** is usable to manufacture an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(242) In comparison with layout design **1200** of FIG. **12**, layout design **1400** includes a second conductive layout pattern **1401a** instead of second conductive layout pattern **1201a** and a third conductive layout pattern **1401b** instead of third conductive layout pattern **1201b**.

(243) Second conductive layout pattern **1401a** is a variation of second conductive layout pattern **401b**, and is usable to manufacture second rail **102b** (FIGS. 1A-1B, 2A-2C). Second conductive layout pattern **1401a** has a width $W_{sub.3a}$ in the second direction Y.

(244) Third conductive layout pattern **1401b** is a variation of third conductive layout pattern **401c**, and is usable to manufacture third rail **102c** (FIGS. 1A-1B, 2A-2C). Third conductive layout pattern **1401b** has a width $W_{sub.3b}$ in the second direction Y.

(245) By increasing the widths $W_{sub.3a}$, $W_{sub.3b}$ of corresponding second and third conductive feature layout patterns **1401b**, **1401c**, second and third conductive feature layout patterns **1401b**, **1401c** are further exposed to form corresponding conductive structures that provide additional via access points, and additional M1 routing resources than other designs.

(246) In some embodiments, width $W_{sub.3a}$ is the same as width $W_{sub.3b}$. In some embodiments, width $W_{sub.3a}$ differs from width $W_{sub.3b}$. In some embodiments, at least one width of widths $W_{sub.1b}$, $W_{sub.1c}$, $W_{sub.2a}$, and $W_{sub.2b}$ of FIGS. **6-13** is less than at least one width of width $W_{sub.3a}$ and $W_{sub.3b}$.

(247) In FIG. **14**, a center of conductive feature layout pattern **208c'** and a center of via layout patterns **206a'**, **206b'**, **214a'**, **402d**, **404d** are aligned with first line **440**. In some embodiments, a center of conductive feature layout pattern **208c'** and a center of each of via layout patterns **206a'**, **206b'**, **214a'**, **402d**, **404d** is aligned with one another in the first direction X.

(248) FIG. **15** is a diagram of a layout design **1500**, in accordance with some embodiments. Layout design **1500** is another variation of layout design **400** or **400'**. Layout design **1500** is usable to manufacture an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(249) In comparison with layout design **1400** of FIG. **14**, layout design **1500** includes a conductive feature layout pattern **1502a** instead of conductive feature layout pattern **208c'** and a conductive feature layout pattern **1502b** instead of conductive feature layout pattern **208b'**.

(250) Conductive feature layout pattern **1502a** is a variation of conductive layout pattern **208c'**, and is usable to manufacture third conductive structure **208c** (FIGS. 1A-1B, 2A-2C). Conductive feature layout pattern **1502a** has a width $W_{sub.4a}$ in the second direction Y.

(251) Conductive feature layout pattern **1502b** is a variation of conductive layout pattern **208b'**, and is usable to manufacture second conductive structure **208b** (FIGS. 1A-1B, 2A-2C). Conductive

feature layout pattern **1502b** has a width $W_{\text{sub.4b}}$ in the second direction Y.

(252) By increasing the widths $W_{\text{sub.3a}}$, $W_{\text{sub.3b}}$ of corresponding second and third conductive feature layout patterns **1401b**, **1401c**, and the widths $W_{\text{sub.4a}}$, $W_{\text{sub.4b}}$ of corresponding conductive feature layout patterns **1502a**, **1502b**, second and third conductive feature layout patterns **1401b**, **1401c** and conductive feature layout patterns **1502a**, **1502b** are further exposed to form corresponding conductive structures that provide additional M0, M2 via access points, and additional M1 routing resources than other designs.

(253) In some embodiments, width $W_{\text{sub.4a}}$ is the same as width $W_{\text{sub.4b}}$. In some embodiments, width $W_{\text{sub.4a}}$ differs from width $W_{\text{sub.4b}}$. In some embodiments, at least one width of widths $W_{\text{sub.1b'}}$ and $W_{\text{sub.1c'}}$ of FIGS. **6-14** & **16-17** is less than at least one width of width $W_{\text{sub.4a}}$ and $W_{\text{sub.4b}}$. In some embodiments, at least one width of widths $W_{\text{sub.3a}}$ or $W_{\text{sub.3b}}$ is greater than or equal to at least one width of width $W_{\text{sub.4a}}$ or $W_{\text{sub.4b}}$.

(254) FIG. **16** is a diagram of a layout design **1600**, in accordance with some embodiments. Layout design **1600** is another variation of layout design **400** or **400'**. Layout design **1600** is usable to manufacture an IC structure similar to IC structure **200** (FIGS. **2A-2C**).

(255) In comparison with layout design **1400** of FIG. **14**, layout design **1600** further includes a conductive feature layout pattern **1602a**, via layout patterns **1604a'**, **1606a'**, **1608a'**, via layout patterns **1604b**, **1606b**, conductive feature layout pattern **1602b**, via layout patterns **1604a**, **1606a**, **1608a**, and via layout patterns **1604b'**, **1606b'**.

(256) Conductive feature layout pattern **1602a** is a variation of conductive layout pattern **208c'**, and is usable to manufacture third conductive structure **208c** (FIGS. **1A-1B**, **2A-2C**). Conductive feature layout pattern **1602a** has a width $W_{\text{sub.5a}}$ in the second direction Y.

(257) Via layout patterns **1604a'**, **1606a'**, **1608a'** are vias **402d**, **206a'**, **214a'**, respectively. Via layout patterns **1604a'**, **1606a'**, **1608a'** are usable to manufacture vias **112a**, **206a**, **214a**, respectively.

(258) Via layout patterns **1604b**, **1606b** are corresponding vias **404d**, **206b'**, and are usable to manufacture corresponding vias **112b**, **206b**.

(259) In FIG. **16**, a center of conductive feature layout pattern **208c'** and a center of via layout patterns **206a'**, **206b'**, **214a'**, **402d**, **404d** are aligned with boundary **910a**. In some embodiments, a center of conductive feature layout pattern **208c'** and a center of each of via layout patterns **206a'**, **206b'**, **214a'**, **402d**, **404d** are aligned with one another in the first direction X. Boundary **910a** is separated from first line **440** in the second direction Y by a distance $D_{\text{sub.6a'}}$.

(260) A center of conductive feature layout pattern **1602a** and a center of via layout patterns **1604a'**, **1606a'**, **1608a'**, **1604b**, **1606b** are aligned with boundary **1610a**. In some embodiments, a center of conductive feature layout pattern **1602a** and a center of each of via layout patterns **1604a'**, **1606a'**, **1608a'**, **1604b**, **1606b** are aligned with one another in the first direction X. Boundary **1610a** is separated from first line **440** in the second direction Y by a distance $D_{\text{sub.8a'}}$. Boundary **1610a** and boundary **910a** are separated from each other in the second direction Y by a distance $D_{\text{sub.9a'}}$.

(261) Conductive feature layout pattern **1602b** is a variation of conductive layout pattern **208b'**, and is usable to manufacture second conductive structure **208b** (FIGS. **1A-1B**, **2A-2C**). Conductive feature layout pattern **1602b** has a width $W_{\text{sub.5b}}$ in the second direction Y.

(262) Via layout patterns **1604b'**, **1606b'** are corresponding vias **406d**, **204a'**, and are usable to manufacture corresponding vias **108a**, **204a**.

(263) Via layout patterns **1604a**, **1606a**, **1608a** are embodiments of corresponding vias **408d**, **204b'**, **212a'**, and are usable to manufacture corresponding vias **108b**, **204b**, **212a**.

(264) In FIG. **16**, a center of conductive feature layout pattern **208b'** and a center of via layout patterns **204a'**, **204b'**, **212a'**, **406d**, **408d** are aligned with boundary **910b**. In some embodiments, a center of conductive feature layout pattern **208b'** and a center of each of via layout patterns **204a'**, **204b'**, **212a'**, **406d**, **408d** are aligned with one another in the first direction X. Boundary **910b** is separated from second line **442** in the second direction Y by a distance $D_{\text{sub.8a}}$.

(265) A center of conductive feature layout pattern **1602b** and a center of via layout patterns **1604a**, **1606a**, **1608a**, **1604b'**, **1606b'** are aligned with boundary **1610b**. In some embodiments, a center of conductive feature layout pattern **1602b** and a center of each of via layout patterns **1604a**, **1606a**, **1608a**, **1604b'**, **1606b'** are aligned with one another in the first direction X. Boundary **1610b** is separated from second line **442** in the second direction Y by a distance $D_{sub.6a}$. Boundary **1610b** and boundary **910b** are separated from each other in the second direction Y by a distance $D_{sub.9a}$.

(266) By increasing the widths $W_{sub.3a}$, $W_{sub.3b}$ of corresponding second and third conductive feature layout patterns **1401b**, **1401c**, second and third conductive feature layout patterns **1401b**, **1401c** are further exposed to form corresponding conductive structures that provide additional M0 via access points. By adding conductive feature layout patterns **1602a**, **1602b**, and separating conductive feature layout patterns **1602a**, **1602b** from corresponding conductive feature layout patterns **208c'**, **208b'** in the second direction Y to form corresponding conductive structures that provide additional M2 via access points than other designs, without also reducing M0 via access points by not covering the center portion of second and third conductive feature layout patterns **1401b**, **1401c**.

(267) In some embodiments, a width of widths $W_{sub.1b'}$, $W_{sub.1c'}$, $W_{sub.5a}$, or $W_{sub.5b}$ is the same as a different width of widths $W_{sub.1b'}$, $W_{sub.1c'}$, $W_{sub.5a}$, or $W_{sub.5b}$. In some embodiments, a width of widths $W_{sub.3a}$ or $W_{sub.3b}$ is greater than or equal to a width of widths $W_{sub.1b'}$, $W_{sub.1c'}$, $W_{sub.5a}$, or $W_{sub.5b}$.

(268) In some embodiments, distance $D_{sub.9a}$ is the same as distance $D_{sub.9a}$. In some embodiments, distance $D_{sub.9a}$ differs from distance $D_{sub.9a}$. In some embodiments, a distance of distances $D_{sub.6a}$, $D_{sub.8a}$, $D_{sub.6a'}$ or $D_{sub.8a'}$ is the same as a different distance of distances $D_{sub.6a}$, $D_{sub.8a}$, $D_{sub.6a'}$ or $D_{sub.8a'}$. In some embodiments, a distance of distances $D_{sub.6a}$, $D_{sub.8a}$, $D_{sub.6a'}$ or $D_{sub.8a'}$ differs from a different distance of distances $D_{sub.6a}$, $D_{sub.8a}$, $D_{sub.6a'}$ or $D_{sub.8a'}$.

(269) FIG. 17 is a diagram of a layout design **1700**, in accordance with some embodiments.

(270) Layout design **1700** is another variation of layout design **400** or **400'**. Layout design **1700** is usable to manufacture an IC structure similar to IC structure **200** (FIGS. 2A-2C).

(271) In comparison with layout design **1400** of FIG. 14, layout design **1700** includes power layout patterns **402'**, **404'**, **406'** and **408'** instead of corresponding power layout patterns **402**, **404**, **406** and **408**. Power layout patterns **402'**, **404'** are asymmetric with respect to first line **440**, and power layout patterns **406'**, **408'** are asymmetric with respect to second line **442**.

(272) Power layout patterns **402'**, **404'**, **406'** and **408'** of layout pattern **1700** include corresponding conductive feature layout patterns **402a'**, **404a'**, **406a'** and **408a'** instead of corresponding conductive feature layout patterns **402a**, **404a**, **406a** and **408a** shown in FIG. 14.

(273) By utilizing power layout patterns **402'**, **404'**, **406'**, **408'**, conductive feature layout patterns **402a'**, **404a'**, **406a'**, **408a'** are exposed to form corresponding conductive structures that provide additional M1 via access points and additional M1 routing resources, and second and third conductive feature layout patterns **1401b**, **1401c** are further exposed to form corresponding conductive structures that provide additional M0 via access points than other designs. By increasing the widths $W_{sub.3a}$, $W_{sub.3b}$ of corresponding second and third conductive feature layout patterns **1401b**, **1401c**, to form corresponding conductive structures provide additional M0 via access points.

(274) A length of a conductive feature layout pattern of the conductive feature layout patterns **402a'**, **404a'**, **406a'** or **408a'** is greater than a length of a corresponding conductive feature layout pattern of the conductive feature layout patterns **402a**, **404a**, **406a** or **408a** shown in FIG. 14.

(275) In some embodiments, a length of conductive feature layout patterns **602a** or **602i** of FIG. 17 is less than one or more of conductive feature layout patterns **602b**, **602c**, . . . , or **602h**. In some embodiments, a length of conductive feature layout patterns **602a** or **602i** of layout pattern **1700** is less than a length of conductive feature layout patterns **602a** or **602i** of layout pattern **1400**.

(276) FIG. 18 is a diagram of a layout design 1800, in accordance with some embodiments. Layout design 1800 is another variation of layout design 400 or 400'. Layout design 1800 is usable to manufacture an IC structure similar to IC structure 200 (FIGS. 2A-2C).

(277) In comparison with layout design 1700 of FIG. 17, layout design 1800 includes conductive feature layout pattern 1502a instead of conductive feature layout pattern 208c' and conductive feature layout pattern 1502b instead of conductive feature layout pattern 208b'. In some embodiments, at least one width of widths W.sub.1b' or W.sub.1c' of FIG. 17 is less than at least one width of width W.sub.4a or W.sub.4b. In some embodiments, at least one width of widths W.sub.3a or W.sub.3b is greater than or equal to at least one width of width W.sub.4a or W.sub.4b.

(278) By increasing the widths W.sub.3a, W.sub.3b of corresponding second and third conductive feature layout patterns 1401b, 1401c, and the widths W.sub.4a, W.sub.4b of corresponding conductive feature layout patterns 1502a, 1502b, second and third conductive feature layout patterns 1401b, 1401c and conductive feature layout patterns 1502a, 1502b are further exposed to form corresponding conductive structures that provide additional M0, M2 via access points, and additional M1 routing resources than other designs.

(279) In some embodiments, layout design 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700 or 1800 is a layout design of first region 420, 422, 424 or 426 (FIGS. 4A-4B or 5A-5B). In some embodiments, layout design 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700 or 1800 is a layout design of a standard cell.

(280) By providing additional via access points or additional M1 routing resources, one or more layout designs 600-1800 occupy less area than other designs, and one or more IC structures 100, 100', 200, 300 occupy less area than other IC structures. Other configurations of via layout patterns and corresponding via layers or conductive feature layout patterns and corresponding metal layers are within the scope of the present disclosure.

(281) FIG. 19A is a flowchart of a method 1900 of forming or manufacturing an IC in accordance with some embodiments. It is understood that additional operations may be performed before, during, and/or after the method 1900 depicted in FIG. 19, and that some other processes may only be briefly described herein. In some embodiments, the method 1900 is usable to form integrated circuits, such as IC structure 100, 100', 200 or 300 (FIGS. 1A-1B, 2A-2C or 3A-3C). In some embodiments, the method 1900 is usable to form integrated circuits having similar structural relationships as one or more of layout patterns 400-1800 (FIGS. 4A-4B, 5A-5B or 6-18).

(282) In operation 1902 of method 1900, a set of rails 120 are formed. In some embodiments, the set of rails 120 includes first rail 102a, second rail 102b or third rail 102c. In some embodiments, the set of rails 120 extends in the first direction X and is located at a first level. In some embodiments, the first level is M0. Other configurations of via levels or metal levels are within the scope of the present disclosure. In some embodiments, each rail of the set of rails 120 is separated from one another in second direction Y. In some embodiments, operation 1902 of method 1900 includes at least operation 1920 or 1922 of method 1900' (shown below).

(283) In operation 1904, a first set of conductive structures 130 are formed. In some embodiments, the first set of conductive structures 130 includes one or more of conductive structures 106a, 106b, . . . , 106f, conductive structures 110a, 110b, . . . , 110f, or conductive structures 114a, 114b, . . . , 114f. In some embodiments, the first set of conductive structures 130 extends in the second direction Y, overlaps the set of rails 120 and is located at a second level different than the first level. In some embodiments, the second level is M1. Other configurations of levels are within the scope of the present disclosure. In some embodiments, the first set of conductive structures 130 includes a first sub-set of conductive structures and a second sub-set of conductive structures. In some embodiments, the first sub-set of conductive structures includes conductive structures 106a, 106b, . . . , 106f, and the second sub-set of conductive structures includes conductive structures 114a, 114b, . . . , 114f. In some embodiments, at least one conductive structure of the first sub-set of conductive structures is aligned with at least one conductive structure of the second sub-set of

conductive structures in the second direction. In some embodiments, operation **1904** of method **1900** includes at least operation **1924** or **1926** of method **1900'** (shown below).

(284) In operation **1906**, a first set of vias **125** is formed between the set of rails **120** and the first set of conductive structures **130**. In some embodiments, the first set of vias **125** includes one or more of vias **104a**, **104b**, . . . , **104f**, vias **108a**, **108b**, . . . , **108f** or vias **112a**, **112b**, . . . , **112f**. In some embodiments, each via of the first set of vias **125** is located where each conductive structure of the first set of conductive structures **130** overlaps each rail of the set of rails **120**. In some embodiments, the first set of vias **125** couple the set of rails **120** to the first set of conductive structures **130**. In some embodiments, operation **1906** of method **1900** includes at least operation **1928** or **1930** of method **1900'** (shown below). In some embodiments, the first set of vias **125** are located at V0. In some embodiments, the first set of vias **125** are located between M0 and M1. Other configurations of via levels or metal levels are within the scope of the present disclosure.

(285) In operation **1908**, a second set of conductive structures **230** is formed over the set of rails **120**. In some embodiments, the second set of conductive structures **230** includes first conductive structure **208a**, second conductive structure **208b** or third conductive structure **208c**. In some embodiments, the second set of conductive structures **230** extends in the first direction X, and is located at a third level different than the first level and the second level. In some embodiments, the third level is M2. Other configurations of levels are within the scope of the present disclosure. In some embodiments, each conductive structure of the second set of conductive structures **230** is separated from one another in the second direction Y. In some embodiments, operation **1908** of method **1900** includes at least operation **1932** or **1934** of method **1900'** (shown below).

(286) In operation **1910**, a second set of vias **225** is formed between the second set of conductive structures **230** and the first set of conductive structures **130**. In some embodiments, the second set of vias **225** includes one or more of vias **202a**, **202b**, . . . , **202f**, vias **204a**, **204b**, . . . , **204f** or vias **206a**, **206b**, . . . , **206f**. In some embodiments, the second set of vias **225** couple the second set of conductive structures **230** to the first set of conductive structures **130**. In some embodiments, each via of the second set of vias **225** is located where each conductive structure of the second set of conductive structures **230** overlaps each conductive structure of the first set of conductive structures **130**. In some embodiments, operation **1910** of method **1900** includes at least operation **1936** or **1938** of method **1900'** (shown below). In some embodiments, the second set of vias **225** are located at V1. In some embodiments, the second set of vias **225** are located between M1 and M2. Other configurations of via levels or metal levels are within the scope of the present disclosure.

(287) In operation **1912**, a third set of conductive structures **240** is formed. In some embodiments, the third set of conductive structures **240** includes one or more of conductive structures **220a**, **220b**, . . . , **220f**. In some embodiments, the third set of conductive structures **240** extends in the second direction Y, and overlaps the set of rails **120** and the first set of conductive structures **130**. In some embodiments, the third set of conductive structures **240** is located at a fourth level different than the first level, the second level and the third level. In some embodiments, the fourth level is M3. Other configurations of levels are within the scope of the present disclosure. In some embodiments, each conductive structure of the third set of conductive structures **240** is separated from one another in the first direction X. In some embodiments, operation **1912** of method **1900** includes at least operation **1940** or **1942** of method **1900'**.

(288) In operation **1914**, a third set of vias **235** is formed between the third set of conductive structures **240** and the second set of conductive structures **230**. In some embodiments, the third set of vias **235** includes one or more of vias **210a**, **210b**, **210c**, vias **212a**, **212b**, **212c** or vias **214a**, **214b**, **214c**. In some embodiments, the third set of vias **235** couple the third set of conductive structures **240** to the second set of conductive structures **230**. In some embodiments, operation **1914** of method **1900** includes at least operation **1944** or **1946** of method **1900'** (shown below). In some embodiments, the third set of vias **235** are located at V2. In some embodiments, the third set

of vias **235** are located between **M2** and **M3** (not shown). Other configurations of via levels or metal levels are within the scope of the present disclosure. In some embodiments, method **1900** further includes one or more operations to manufacture a set of masks based on one or more layout patterns of method **1900'** (shown below), and using the set of masks to manufacture the one or more structural elements in method **1900**.

(289) FIG. **19B** is a flowchart of a method **1900'** of manufacturing an IC in accordance with some embodiments. It is understood that additional operations may be performed before, during, and/or after the method **1900'** depicted in FIG. **19B**, and that some other processes may only be briefly described herein. In some embodiments, the method **1900'** is usable to form ICs, such as IC structure **100**, **100'**, **200** or **300** (FIGS. **1A-1B**, **2A-2C** or **3A-3C**). In some embodiments, the method **1900'** is usable to form integrated circuits having similar structural relationships as one or more of layout patterns **400-1800** (FIGS. **4A-4B**, **5A-5B** or **6-18**).

(290) In operation **1920** of method **1900'**, a first set of conductive structure layout patterns are generated. The first set of conductive structure layout patterns correspond to fabricating a set of rails **120** of IC structure **100**, **100'**, **200** or **300**. In some embodiments, the first set of conductive structure layout patterns includes one or more of first conductive layout pattern **401a**, second conductive layout pattern **401b**, third conductive layout pattern **401c**, second conductive layout pattern **1201a**, third conductive layout pattern **1201b**, second conductive layout pattern **1401a** or third conductive layout pattern **1401b**. In some embodiments, the set of rails **120** includes first rail **102a**, second rail **102b** or third rail **102c**. The first set of conductive structure layout patterns extend in a first direction **X**. In some embodiments, each conductive structure layout pattern of the first set of conductive structure layout patterns is separated from one another in a second direction **Y**.

(291) In operation **1922** of method **1900'**, the first set of conductive structure layout patterns are placed on a first layout level. In some embodiments, operation **1922** further includes manufacturing a set of rails **120** based on the first set of conductive structure layout patterns. In some embodiments, the set of rails **120** includes first rail **102a**, second rail **102b** or third rail **102c**. In some embodiments, the first layout level is **M0** of IC structure **100**, **200**, **300**. Other configurations of levels are within the scope of the present disclosure.

(292) In operation **1924**, a second set of conductive structure layout patterns is generated. The second set of conductive structure layout patterns correspond to fabricating a first set of conductive structures **130** of IC structure **100**, **100'**, **200** or **300**. In some embodiments, the second set of conductive structure layout patterns includes one or more of conductive feature layout pattern **402a**, **404a**, **406a**, **408a**, **410a**, **412a**, **402a'**, **404a'**, **406a'**, **408a'**, **410a'**, **412a'**, **702a**, **704a**, **802a**, **804a**, **806a** or **808a**. In some embodiments, the second set of conductive structure layout patterns includes one or more of power layout patterns **402**, **404**, **406**, **408**, **410**, **412**, **402'**, **404'**, **406'**, **408'**, **410'**, **412'**, **702**, **704**, **802**, **804**, **806** or **808**. In some embodiments, the first set of conductive structures **130** includes one or more of conductive structures **106a**, **106b**, . . . , **106f**, conductive structures **110a**, **110b**, . . . , **110f**, or conductive structures **114a**, **114b**, . . . , **114f**. In some embodiments, the second set of conductive structure layout patterns extend in the second direction **Y** and overlap the first set of conductive structure layout patterns.

(293) In operation **1926**, the second set of conductive structure layout patterns is placed on a second layout level different than the first layout level. In some embodiments, operation **1926** further includes manufacturing a first set of conductive structures **130** based on the second set of conductive structure layout patterns. In some embodiments, the first set of conductive structures **130** includes one or more of conductive structures **106a**, **106b**, . . . , **106f**, conductive structures **110a**, **110b**, . . . , **110f**, or conductive structures **114a**, **114b**, . . . , **114f**. In some embodiments, the second layout level is **M1** of IC structure **100**, **200**, **300**. Other configurations of levels are within the scope of the present disclosure.

(294) In operation **1928**, a first set of via layout patterns are generated. The first set of via layout patterns correspond to fabricating a first set of vias **125** of IC structure **100**, **100'**, **200** or **300**. In

some embodiments, the first set of via layout patterns includes one or more of via layout pattern **402d**, **404d**, **406d**, **408d**, **410d**, **412d**, **702d**, **704d**, **802d**, **804d**, **806d**, **808d**, **1604a**, **1604b**, **1604a'** or **1604b'**. In some embodiments, the first set of vias **125** includes one or more of vias **104a**, **104b**, . . . , **104f**, vias **108a**, **108b**, . . . , **108f** or vias **112a**, **112b**, . . . , **112f**.

(295) In operation **1930**, the first set of via layout patterns are placed between the first layout level and the second layout level. In some embodiments, operation **1930** further includes manufacturing a first set of vias **125** based on the first set of via layout patterns. In some embodiments, the first set of vias **125** includes one or more of vias **104a**, **104b**, . . . , **104f**, vias **108a**, **108b**, . . . , **108f** or vias **112a**, **112b**, . . . , **112f**. In some embodiments, each via layout pattern of the first set of via layout patterns is located where each conductive structure layout pattern of the second set of conductive structure layout patterns overlaps each conductive structure layout pattern of the first set of conductive structure layout patterns. In some embodiments, the first set of via layout patterns couple the first set of conductive structure layout patterns to the second set of conductive structure layout patterns. Other configurations of levels are within the scope of the present disclosure.

(296) In operation **1932**, a third set of conductive structure layout patterns are generated. The third set of conductive structure layout patterns correspond to fabricating a second set of conductive structures **230** of IC structure **100**, **100'**, **200** or **300**. In some embodiments, the third set of conductive structure layout patterns includes one or more of conductive feature layout pattern **208b'**, **208c'**, **1502a**, **1502b**, **1602a** or **1602b**. In some embodiments, the second set of conductive structures **230** includes one or more of first conductive structure **208a**, second conductive structure **208b** or third conductive structure **208c**. In some embodiments, the third set of conductive structure layout patterns extend in the first direction X and is directly over the first set of conductive structure layout patterns. In some embodiments, each conductive structure layout pattern of the third set of conductive structure layout patterns is separated from one another in the second direction Y.

(297) In operation **1934**, the third set of conductive structure layout patterns are placed on a third layout level. In some embodiments, operation **1934** further includes manufacturing a second set of conductive structures **230** based on the third set of conductive structure layout patterns. In some embodiments, the second set of conductive structures **230** includes first conductive structure **208a**, second conductive structure **208b** or third conductive structure **208c**. In some embodiments, the third layout level is M2 of IC structure **100**, **200**, **300**. Other configurations of levels are within the scope of the present disclosure.

(298) In operation **1936**, a second set of via layout patterns are generated. The second set of via layout patterns correspond to fabricating a second set of vias **225** of IC structure **100**, **100'**, **200** or **300**. In some embodiments, the second set of via layout patterns includes one or more of **204a'**, **204b'**, **204c'**, **204d'**, **204e'**, **206a'**, **206b'**, **206c'**, **206d'**, **206e'**, **1606a**, **1606b**, **1606a'** or **1606b'**. In some embodiments, the second set of vias **225** includes one or more of vias **202a**, **202b**, . . . , **202f**, vias **204a**, **204b**, . . . , **204f** or vias **206a**, **206b**, . . . , **206f**.

(299) In operation **1938**, the second set of via layout patterns is placed between the second layout level and the third layout level. In some embodiments, operation **1938** further includes manufacturing a second set of vias **225** based on the second set of via layout patterns. In some embodiments, the second set of vias **225** includes one or more of vias **202a**, **202b**, . . . , **202f**, vias **204a**, **204b**, . . . , **204f** or vias **206a**, **206b**, . . . , **206f**. In some embodiments, the second set of via layout patterns couple the third set of conductive structure layout patterns to the second set of conductive structure layout patterns. In some embodiments, each via layout pattern of the second set of via layout patterns is located where each conductive structure layout pattern of the third set of conductive structure layout patterns overlaps each conductive structure layout pattern of the second set of conductive structure layout patterns. Other configurations of levels are within the scope of the present disclosure.

(300) In operation **1940**, a fourth set of conductive structure layout patterns is generated. The

fourth set of conductive structure layout patterns correspond to fabricating a third set of conductive structures **240** of IC structure **100**, **100'**, **200** or **300**. In some embodiments, the fourth set of conductive structure layout patterns includes one or more of conductive feature layout patterns **220a'**, **220b'**, **220c'**, **220d'** or **220e'**. In some embodiments, the fourth set of conductive structures includes one or more of conductive structures **220a**, **220b**, . . . , or **220f**. In some embodiments, the fourth set of conductive structure layout patterns extend in the second direction Y and overlap the first set of conductive structure layout patterns and the second set of conductive structure layout patterns. In some embodiments, each conductive structure layout pattern of the fourth set of conductive structure layout patterns is separated from one another in the first direction X.

(301) In operation **1942**, the fourth set of conductive structure layout patterns is placed on a fourth layout level. In some embodiments, operation **1942** further includes manufacturing a third set of conductive structures **240** based on the fourth set of conductive structure layout patterns. In some embodiments, the third set of conductive structures **240** includes one or more of conductive structures **220a**, **220b**, . . . , **220f**. In some embodiments, the fourth layout level is M3 of IC structure **100**, **200**, **300**. Other configurations of levels are within the scope of the present disclosure.

(302) In operation **1944**, a third set of via layout patterns is generated. The third set of via layout patterns correspond to fabricating a third set of vias **235** of IC structure **100**, **100'**, **200** or **300**. In some embodiments, the third set of via layout patterns includes one or more of **214a'**, **214b'**, **214c'**, **212a'**, **212b'**, **212c'**, **1608a** or **1608a'**. In some embodiments, the third set of vias **235** includes one or more of vias **210a**, **210b**, **210c**, vias **212a**, **212b**, **212c** or vias **214a**, **214b**, **214c**.

(303) In operation **1946**, the third set of via layout patterns is placed between the third layout level and the fourth layout level. In some embodiments, operation **1946** further includes manufacturing a third set of vias **235** based on the third set of via layout patterns. In some embodiments, the third set of vias **235** includes one or more of vias **210a**, **210b**, **210c**, vias **212a**, **212b**, **212c** or vias **214a**, **214b**, **214c**. In some embodiments, the third set of via layout patterns couple the fourth set of conductive structure layout patterns to the third set of conductive structure layout patterns. Other configurations of levels are within the scope of the present disclosure.

(304) In some embodiments, one or more of operations **1920**, **1922**, **1924**, **1926**, **1928**, **1930**, **1932**, **1934**, **1936**, **1938**, **1940**, **1942** or **1944** are optional. In some embodiments, method **1900'** further includes one or more operations to manufacture a set of masks based on one or more layout patterns of method **1900'**. In these embodiments, method **1900'** further includes one or more operations to manufacture integrated circuit IC structure **100**, **100'**, **200** or **300** using the set of masks.

(305) One or more of operations **1920**, **1922**, **1924**, **1926**, **1928**, **1930**, **1932**, **1934**, **1936**, **1938**, **1940**, **1942**, **1944** or **1946** is performed by a processing device configured to execute instructions for manufacturing an IC, such as IC structure **100**, **100'**, **200**, or **300**. In some embodiments, one or more of operations **1920**, **1922**, **1924**, **1926**, **1928**, **1930**, **1932**, **1934**, **1936**, **1938**, **1940**, **1942**, **1944** or **1946** is performed using a same processing device as that used in a different one or more of operations **1920**, **1922**, **1924**, **1926**, **1928**, **1930**, **1932**, **1934**, **1936**, **1938**, **1940**, **1942**, **1944** or **1946**. In some embodiments, a different processing device is used to perform one or more of operations **1920**, **1922**, **1924**, **1926**, **1928**, **1930**, **1932**, **1934**, **1936**, **1938**, **1940**, **1942**, **1944** or **1946** from that used to perform a different one or more of operations **1920**, **1922**, **1924**, **1926**, **1928**, **1930**, **1932**, **1934**, **1936**, **1938**, **1940**, **1942**, **1944** or **1946**.

(306) FIG. **20** is a schematic view of a system **2000** for designing an IC layout design in accordance with some embodiments. System **2000** includes a hardware processor **2002** and a non-transitory, computer readable storage medium **2004** encoded with, i.e., storing, the computer program code **2006**, i.e., a set of executable instructions. Computer readable storage medium **2004** is also encoded with instructions **2007** for interfacing with manufacturing machines for producing the integrated circuit. The processor **2002** is electrically coupled to the computer readable storage

medium **2004** via a bus **2008**. The processor **2002** is also electrically coupled to an I/O interface **2010** by bus **2008**. A network interface **2012** is also electrically connected to the processor **2002** via bus **2008**. Network interface **2012** is connected to a network **2014**, so that processor **2002** and computer readable storage medium **2004** are capable of connecting to external elements via network **2014**. The processor **2002** is configured to execute the computer program code **2006** encoded in the computer readable storage medium **2004** in order to cause system **2000** to be usable for performing a portion or all of the operations as described in method **1900'**.

(307) In some embodiments, the processor **2002** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

(308) In some embodiments, the computer readable storage medium **2004** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, the computer readable storage medium **2004** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In some embodiments using optical disks, the computer readable storage medium **2004** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

(309) In some embodiments, the storage medium **2004** stores the computer program code **2006** configured to cause system **2000** to perform method **1900** or **1900'**. In some embodiments, the storage medium **2004** also stores information needed for performing method **1900** or **1900'** as well as information generated during performing method **1900** or **1900'**, such as layout design **2016**, first set of conductive feature layout patterns **2018**, second set of conductive feature layout patterns **2020**, third set of conductive feature layout patterns **1222**, fourth set of conductive feature layout patterns **2024**, fifth set of conductive feature layout patterns **2026**, first set of via layout patterns **2028**, second set of via layout patterns **2030**, third set of via layout patterns **2032** and user interface **2034**, and/or a set of executable instructions to perform the operation of method **1900** or **1900'**.

(310) In some embodiments, the storage medium **2004** stores instructions **2007** for interfacing with manufacturing machines. The instructions **2007** enable processor **2002** to generate manufacturing instructions readable by the manufacturing machines to effectively implement method **1900** or **1900'** during a manufacturing process.

(311) System **2000** includes I/O interface **2010**. I/O interface **2010** is coupled to external circuitry. In some embodiments, I/O interface **2010** includes a keyboard, keypad, mouse, trackball, trackpad, and/or cursor direction keys for communicating information and commands to processor **2002**.

(312) System **2000** also includes network interface **2012** coupled to the processor **2002**. Network interface **2012** allows system **2000** to communicate with network **2014**, to which one or more other computer systems are connected. Network interface **2012** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interface such as ETHERNET, USB, or IEEE-1394. In some embodiments, method **1900** or **1900'** is implemented in two or more systems **2000**, and information such as layout design, first set of conductive feature layout patterns, second set of conductive feature layout patterns, third set of conductive feature layout patterns, fourth set of conductive feature layout patterns, fifth set of conductive feature layout patterns, first set of via layout patterns, second set of via layout patterns, third set of via layout patterns and user interface are exchanged between different systems **2000** by network **2014**.

(313) System **2000** is configured to receive information related to a layout design through I/O interface **2010** or network interface **2012**. The information is transferred to processor **2002** by bus **2008** to determine a layout design for producing integrated circuit structure **100**, **100'**, **200** or **300**. The layout design is then stored in computer readable medium **2004** as layout design **2016**. System **2000** is configured to receive information related to a first set of conductive feature layout patterns through I/O interface **2010** or network interface **2012**. The information is stored in computer

readable medium **2004** as first set of conductive feature layout patterns **2018**. System **2000** is configured to receive information related to a second set of conductive feature layout patterns through I/O interface **2010** or network interface **2012**. The information is stored in computer readable medium **2004** as second set of conductive feature layout patterns **2020**. System **2000** is configured to receive information related to a third set of conductive feature layout patterns through I/O interface **2010** or network interface **2012**. The information is stored in computer readable medium **2004** as third set of conductive feature layout patterns **2022**. System **2000** is configured to receive information related to a fourth set of conductive feature layout patterns through I/O interface **2010** or network interface **2012**. The information is stored in computer medium **2004** as fourth set of conductive feature layout patterns **2024**. System **2000** is configured to receive information related to a fifth set of conductive feature layout patterns through I/O interface **2010** or network interface **2012**. The information is stored in computer readable medium **2004** as fifth set of conductive feature layout patterns **2026**. System **2000** is configured to receive information related to a first set of via layout patterns through I/O interface **2010** or network interface **2012**. The information is stored in computer readable medium **2004** as first set of via layout patterns **2028**. System **2000** is configured to receive information related to a second set of via layout patterns through I/O interface **2010** or network interface **2012**. The information is stored in computer readable medium **2004** as second set of via layout patterns **2030**. System **2000** is configured to receive information related to a third set of via layout patterns through I/O interface **2010** or network interface **2012**. The information is stored in computer readable medium **2004** as third set of via layout patterns **2032**. System **2000** is configured to receive information related to a user interface through I/O interface **2010** or network interface **2012**. The information is stored in computer readable medium **2004** as user interface **2034**.

(314) In some embodiments, method **1900** or **1900'** is implemented as a standalone software application for execution by a processor. In some embodiments, method **1900** or **1900'** is implemented as a software application that is a part of an additional software application. In some embodiments, method **1900** or **1900'** is implemented as a plug-in to a software application. In some embodiments, method **1900** or **1900'** is implemented as a software application that is a portion of an EDA tool. In some embodiments, method **1900** or **1900'** is implemented as a software application that is used by an EDA tool. In some embodiments, the EDA tool is used to generate a layout of the integrated circuit device. In some embodiments, the layout is stored on a non-transitory computer readable medium. In some embodiments, the layout is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool. In some embodiments, the layout is generated based on a netlist which is created based on the schematic design. In some embodiments, method **1900** is implemented by a manufacturing device to manufacture an integrated circuit (e.g., integrated circuit **100**, **200** or **300**) using a set of masks manufactured based on one or more layout designs (e.g., layout design **400**, **400'**, **500**, **500'**, **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**, **1400**, **1500**, **1600**, **1700** or **1800**) generated by system **2000**.

(315) System **2000** of FIG. **20** generates layout designs (e.g., layout design **400**, **400'**, **500**, **500'**, **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**, **1400**, **1500**, **1600**, **1700** or **1800**) of integrated circuit structure **100**, **100'**, **200** or **300** that occupy less area and provide better routing resources than other approaches.

(316) One aspect of this description relates to an integrated circuit structure. In some embodiments, the integrated circuit structure includes a first power rail extending in a first direction and being located at a first level. In some embodiments, the integrated circuit structure further includes a second power rail extending in the first direction and being located at the first level, the second power rail being separated from the first power rail in a second direction different from the first direction. In some embodiments, the integrated circuit structure further includes a first set of conductive structures extending in the second direction, over the first power rail and being located

at a second level different than the first level. In some embodiments, the integrated circuit structure further includes a second set of conductive structures extending in the second direction, over the second power rail, being located at the second level, and being separated from the first set of conductive structures in the second direction. In some embodiments, the integrated circuit structure further includes a first conductive structure extending in the second direction, overlapping a first conductive structure of the first set of conductive structures and a first conductive structure of the second set of conductive structures, and being located at a third level different than the first level and the second level. In some embodiments, the integrated circuit structure further includes a second conductive structure extending in the second direction, overlapping a second conductive structure of the first set of conductive structures and a second conductive structure of the second set of conductive structures, being located at the third level, and being separated from the first conductive structure in the first direction. In some embodiments, the integrated circuit structure further includes a third conductive structure extending in the second direction, overlapping a third conductive structure of the first set of conductive structures and a third conductive structure of the second set of conductive structures, being located at the third level, and being separated from the first conductive structure in the first direction.

(317) Another aspect of this description relates an integrated circuit structure. In some embodiments, the integrated circuit structure includes a first power rail extending in a first direction, being located at a first level, and configured to provide a first supply voltage. In some embodiments, the integrated circuit structure further includes a second power rail extending in the first direction, being located at the first level, and configured to provide a second supply voltage different from the first supply voltage, the second power rail being separated from the first power rail in a second direction different from the first direction. In some embodiments, the integrated circuit structure further includes a first set of conductive structures extending in the second direction, over the first power rail and being located at a second level different than the first level. In some embodiments, the integrated circuit structure further includes a second set of conductive structures extending in the second direction, over the second power rail, being located at the second level, and being separated from the first set of conductive structures in at least the first direction or the second direction, the first set of conductive structures and the second set of conductive structures are not aligned with each other in the second direction. In some embodiments, the integrated circuit structure further includes a first conductive structure extending in the first direction, covering a portion of the first power rail, overlapping the first set of conductive structures, and being located at a third level different than the first level and the second level. In some embodiments, the integrated circuit structure further includes a second conductive structure extending in the first direction, covering a portion of the second power rail, overlapping the second set of conductive structures, being located at the third level, and being separated from the first conductive structure in the second direction.

(318) Still another aspect of this description relates to a method of forming an integrated circuit structure. In some embodiments, the method includes generating a set of power rail patterns corresponding to a set of power rails, the set of power rail patterns being on a first level, the set of power rail patterns extending in a first direction, each pattern of the set of power rail patterns being separated from one another in a second direction different from the first direction. In some embodiments, the method further includes generating a first set of conductive patterns corresponding to a first set of conductors, the first set of conductive patterns being on a second level different from the first level, the first set of conductive patterns extending in the second direction and overlapping the set of power rail patterns. In some embodiments, the method further includes generating a second set of conductive patterns corresponding to a second set of conductors, the second set of conductive patterns being on the first level and between the set of power rail patterns, the second set of conductive patterns extending in the first direction. In some embodiments, the method further includes generating a third set of conductive patterns

corresponding to a third set of conductors, the third set of conductive patterns being on the second level and between the set of power rail patterns, the third set of conductive patterns extending in the second direction. In some embodiments, the method further includes generating a fourth set of conductive patterns corresponding to a fourth set of conductors, the fourth set of conductive patterns being on a third level different from the first level and the second level, the fourth set of conductive patterns extending in the first direction, and covering a portion of the first set of conductive patterns, each conductive pattern of the fourth set of conductive patterns being separated from one another in the second direction. In some embodiments, the method further includes manufacturing the integrated circuit structure based on at least one of the above patterns. (319) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated circuit structure comprising: a first power rail extending in a first direction and being located at a first level; a second power rail extending in the first direction and being located at the first level, the second power rail being separated from the first power rail in a second direction different from the first direction; a first set of conductive structures extending in the second direction, over the first power rail and being located at a second level different than the first level; a second set of conductive structures extending in the second direction, over the second power rail, being located at the second level, and being separated from the first set of conductive structures in the second direction; a first conductive structure extending in the second direction, overlapping a first conductive structure of the first set of conductive structures and a first conductive structure of the second set of conductive structures, and being located at a third level different than the first level and the second level; a second conductive structure extending in the second direction, overlapping a second conductive structure of the first set of conductive structures and a second conductive structure of the second set of conductive structures, being located at the third level, and being separated from the first conductive structure in the first direction; and a third conductive structure extending in the second direction, overlapping a third conductive structure of the first set of conductive structures and a third conductive structure of the second set of conductive structures, being located at the third level, and being separated from the first conductive structure in the first direction.
2. The integrated circuit structure of claim 1, further comprising: a fourth conductive structure extending in the first direction, being over the first power rail and the first set of conductive structures, and being located at a fourth level different than the first level, the second level and the third level; and a fifth conductive structure extending in the first direction, being over the second power rail and the second set of conductive structures, and being located at the fourth level, and being separated from the fourth conductive structure in the second direction.
3. The integrated circuit structure of claim 2, further comprising: a first set of vias coupling the first power rail and the first set of conductive structures to each other; and a second set of vias coupling the second power rail and the second set of conductive structures to each other.
4. The integrated circuit structure of claim 3, further comprising: a third set of vias between the fourth conductive structure and the first set of conductive structures, and the third set of vias coupling the fourth conductive structure to the first set of conductive structures; and a fourth set of

- vias between the fifth conductive structure and the second set of conductive structures, and the fourth set of vias coupling the fifth conductive structure to the second set of conductive structures.
5. The integrated circuit structure of claim 4, further comprising: a fifth set of vias between the first conductive structure and the fourth conductive structure, and the fifth set of vias coupling the first conductive structure to the fourth conductive structure; and a sixth set of vias between the second conductive structure and the fifth conductive structure, and the sixth set of vias coupling the second conductive structure to the fifth conductive structure.
6. The integrated circuit structure of claim 2, wherein each of the first power rail and the second power rail have a first width in the second direction; and each of the fourth conductive structure and the fifth conductive structure have a second width in the second direction different than the first width.
7. The integrated circuit structure of claim 1, wherein the first set of conductive structures and the second set of conductive structures are aligned in the second direction.
8. The integrated circuit structure of claim 1, wherein the first set of conductive structures does not overlap the second power rail; and the second set of conductive structures does not overlap the first power rail.
9. The integrated circuit structure of claim 1, further comprising: a third set of conductive structures extending in the second direction, being between the first set of conductive structures and the second set of conductive structures, and being separated from the first set of conductive structures and the second set of conductive structures in the second direction.
10. The integrated circuit structure of claim 9, wherein the third set of conductive structures comprises: a fourth conductive structure having a first length in the second direction, the fourth conductive structure being aligned with the first conductive structure of the first set of conductive structures in the second direction; and a fifth conductive structure having a second length in the second direction, the fifth conductive structure being aligned with the second conductive structure of the first set of conductive structures in the second direction.
11. The integrated circuit structure of claim 10, wherein the first length being different than the second length; or the first length being substantially equal to the second length.
12. An integrated circuit structure comprising: a first power rail extending in a first direction, being located at a first level, and configured to provide a first supply voltage; a second power rail extending in the first direction, being located at the first level, and configured to provide a second supply voltage different from the first supply voltage, the second power rail being separated from the first power rail in a second direction different from the first direction; a first set of conductive structures extending in the second direction, over the first power rail and being located at a second level different than the first level; a second set of conductive structures extending in the second direction, over the second power rail, being located at the second level, and being separated from the first set of conductive structures in at least the first direction or the second direction, the first set of conductive structures and the second set of conductive structures are not aligned with each other in the second direction; a first conductive structure extending in the first direction, covering a portion of the first power rail, overlapping the first set of conductive structures, and being located at a third level different than the first level and the second level; and a second conductive structure extending in the first direction, covering a portion of the second power rail, overlapping the second set of conductive structures, being located at the third level, and being separated from the first conductive structure in the second direction.
13. The integrated circuit structure of claim 12, wherein each conductive structure of the first set of conductive structures is offset in the first direction from a corresponding conductive structure in the second set of conductive structures in the first direction by a first distance.
14. The integrated circuit structure of claim 12, further comprising: a first set of vias coupling the first power rail and the first set of conductive structures to each other; and a second set of vias coupling the second power rail and the second set of conductive structures to each other.

15. The integrated circuit structure of claim 14, further comprising: a third set of vias between the first conductive structure and the first set of conductive structures, and the third set of vias coupling the first conductive structure to the first set of conductive structures; and a fourth set of vias between the second conductive structure and the second set of conductive structures, and the fourth set of vias coupling the second conductive structure to the second set of conductive structures.
16. The integrated circuit structure of claim 15, further comprising: a third set of conductive structures extending in the second direction, overlapping the first power rail, the second power rail, the first set of conductive structures, the first conductive structure and the second conductive structure, and being located at a fourth level different than the first level, the second level and the third level; a fourth set of conductive structures extending in the second direction, overlapping the first power rail, the second power rail, the second set of conductive structures, the first conductive structure and the second conductive structure, being located at the fourth level, and being separated from the third set of conductive structures in the first direction; and a fifth set of conductive structures extending in the second direction, overlapping the first power rail, the second power rail, the first set of conductive structures, the first conductive structure and the second conductive structure, being located at the fourth level, and being separated from the third set of conductive structures and the fourth set of conductive structures in the first direction.
17. The integrated circuit structure of claim 16, further comprising: a fifth set of vias between the third set of conductive structures and the first conductive structure, and the fifth set of vias coupling the third set of conductive structures to the first conductive structure; and a sixth set of vias between the fourth set of conductive structures and the second conductive structure, and the sixth set of vias coupling the fourth set of conductive structures to the second conductive structure.
18. A method of forming an integrated circuit structure, the method comprising: generating a set of power rail patterns corresponding to a set of power rails, the set of power rail patterns being on a first level, the set of power rail patterns extending in a first direction, each pattern of the set of power rail patterns being separated from one another in a second direction different from the first direction; generating a first set of conductive patterns corresponding to a first set of conductors, the first set of conductive patterns being on a second level different from the first level, the first set of conductive patterns extending in the second direction and overlapping the set of power rail patterns; generating a second set of conductive patterns corresponding to a second set of conductors, the second set of conductive patterns being on the first level and between the set of power rail patterns, the second set of conductive patterns extending in the first direction; generating a third set of conductive patterns corresponding to a third set of conductors, the third set of conductive patterns being on the second level and between the set of power rail patterns, the third set of conductive patterns extending in the second direction; generating a fourth set of conductive patterns corresponding to a fourth set of conductors, the fourth set of conductive patterns being on a third level different from the first level and the second level, the fourth set of conductive patterns extending in the first direction, and covering a portion of the first set of conductive patterns, each conductive pattern of the fourth set of conductive patterns being separated from one another in the second direction; and manufacturing the integrated circuit structure based on at least one of the above patterns.
19. The method of claim 18, further comprising: generating a first set of via patterns corresponding to a first set of vias, the first set of via patterns being between the fourth set of conductive patterns and the first set of conductive patterns.
20. The method of claim 19, further comprising: generating a fifth set of conductive patterns corresponding to a fifth set of conductors, the fifth set of conductive patterns being on a fourth level different from the first level, the second level and the third level, the fifth set of conductive patterns extending in the second direction and overlapping the first set of conductive patterns and the fourth set of conductive patterns, and each conductive pattern of the fifth set of conductive patterns being separated from one another in the first direction; and generating a second set of via

patterns corresponding to a second set of vias, the second set of via patterns being between the fifth set of conductive patterns and the fourth set of conductive patterns.
